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Sulfide Remobilization in the Metamorphosed Kayad Sedimentary Exhalative Zn-Pb Deposit, Western India: Evidence from Mode of Occurrence, Texture, Hydrothermal Alteration Features, and Trace Element Chemistry

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Abstract

The Kayad Zn-Pb deposit, situated within the Proterozoic Aravalli-Delhi fold belt in western India, is primarily characterized by sphalerite and galena along with pyrrhotite and chalcopyrite. The mineralization occurs as disseminated ores in quartzite, disseminated/laminated and massive ores in quartz-mica schist, and in pegmatite and quartz veins. The laminated ores conform to the regional schistosity and folding, whereas the massive Zn-Pb ores postdate the pervasive tectonic fabric, accumulating at the fold hinges. The massive ore is characterized by durchbewegung texture, discrete blebs of galena and chalcopyrite in a sphalerite matrix with low interfacial angles, and discrete intergrowths of sulfides and sulfosalts such as pyrargyrite, gudmundite, Ag-tetrahedrite, and breithauptite. Geochemical analyses of sulfides reveal microinclusions of sulfosalts comprising Ag, Sb, Cu, Tl, and As, which are regarded as low-melting chalcophile elements (LMCEs). Hydrothermal alteration is insignificant in the laminated and massive ores but prominent around Fe-Cu ± Zn-Pb and Zn-Pb ± Fe-Cu veins. The alteration assemblages in these veins evince a pervasive K + Na ± Fe alteration, later overprinted by a subsidiary Ca ± Na alteration. We interpret the laminated/disseminated ores to be of syndiagenetic sedimentary-exhalative (SedEx) origin formed within an euxinic basin. Conversely, the textural features, mineralogical composition, lack of associated hydrothermal alterations, and evident structural influence on the emplacement of the massive ores suggest they have been remobilized both via plastic flow and by sulfide partial melting. Temperature estimates of up to 650°C, derived from Ti-in-biotite geothermometry of the metamorphosed host rocks, indicate lower-middle amphibolite facies conditions during regional metamorphism. The initiation of melting at these temperatures was promoted by the desulfurization of pyrite to pyrrhotite in quartz-mica schist, aided by melting point depression due to the presence of LMCEs like Ag, Sb, and As.

Introduction

Sedimentary exhalative (SedEx) deposits, hosted in siliciclastic and carbonate rocks, constitute an important Pb and Zn sulfide resource, accounting for more than 25% of the global production of these metals (Emsbo, 2009). SedEx systems commonly form stratiform to stratabound sulfide ores, primarily of sphalerite and galena, via exhalation of ore-bearing brines that have no direct association with igneous activity onto the sea floor.

The Proterozoic Aravalli-Delhi fold belt, including the adjoining Bhilwara province in western India, is a well-known metallogenic province hosting several Zn-Pb ± Cu ore belts, e.g., Rampura-Agucha, Rajpura-Dariba-Bethumni, Sindesar-Khurd, Zawarmala, and Kayad-Ghugra (Fig. 1). Extensive work on some of these deposits over the last three decades has helped to constrain the nature and timing of mineralization, P-T conditions during deposition, metal sources, and the characteristics of mineralizing fluid (Sarkar and Gupta, 2012 and references therein). These earlier works have interpreted the laminated ores in carbonaceous and dolomitic phyllites, which were remobilized during later tectonothermal events, to be of sedimentary or diagenetic origin (Deb and Pal, 2004; Sarkar and Banerjee, 2004; Shah, 2004). Evidence of anatexis of the original low-grade SedEx ores has been noted at Rampura-Agucha, Rajpura-Dariba, Zawar, and Sindesar-Khurd, with several studies emphasizing the important role of melt-induced mobilization in the formation of massive mineable ores (Frost et al., 2002; Mishra and Bernhardt, 2009; Pruseth et al., 2016; Govindarao et al., 2017; Govindarao et al., 2020; Bhuyan and Hazarika, 2022).

The Kayad deposit is one of the lesser-studied SedEx deposits of the Aravalli-Delhi fold belt. Exploration work in the region was initiated during the late 1980s by the Geological Survey of India (GSI); however, mining operations by Hindustan Zinc Ltd. (HZL) were initiated only in 2013. The ore minerals in the Kayad deposit occur as disseminations/laminations, as massive sulfides, and in pegmatite and quartz veins. The massive sulfide ores are primarily hosted in graphite-bearing quartz-mica schist, and subordinately in the quartz veins, and are being mined for Zn and Pb. However, very little is known about the orebody evolution at Kayad in general and the formation of the mineable ores in particular.

In this study, we integrate evidence from the mode of occurrence, texture, and chemistry of ore and gangue minerals, along with types and patterns of hydrothermal alterations, to decipher the mechanism of formation of the high-grade mineable ores in the Kayad deposit.

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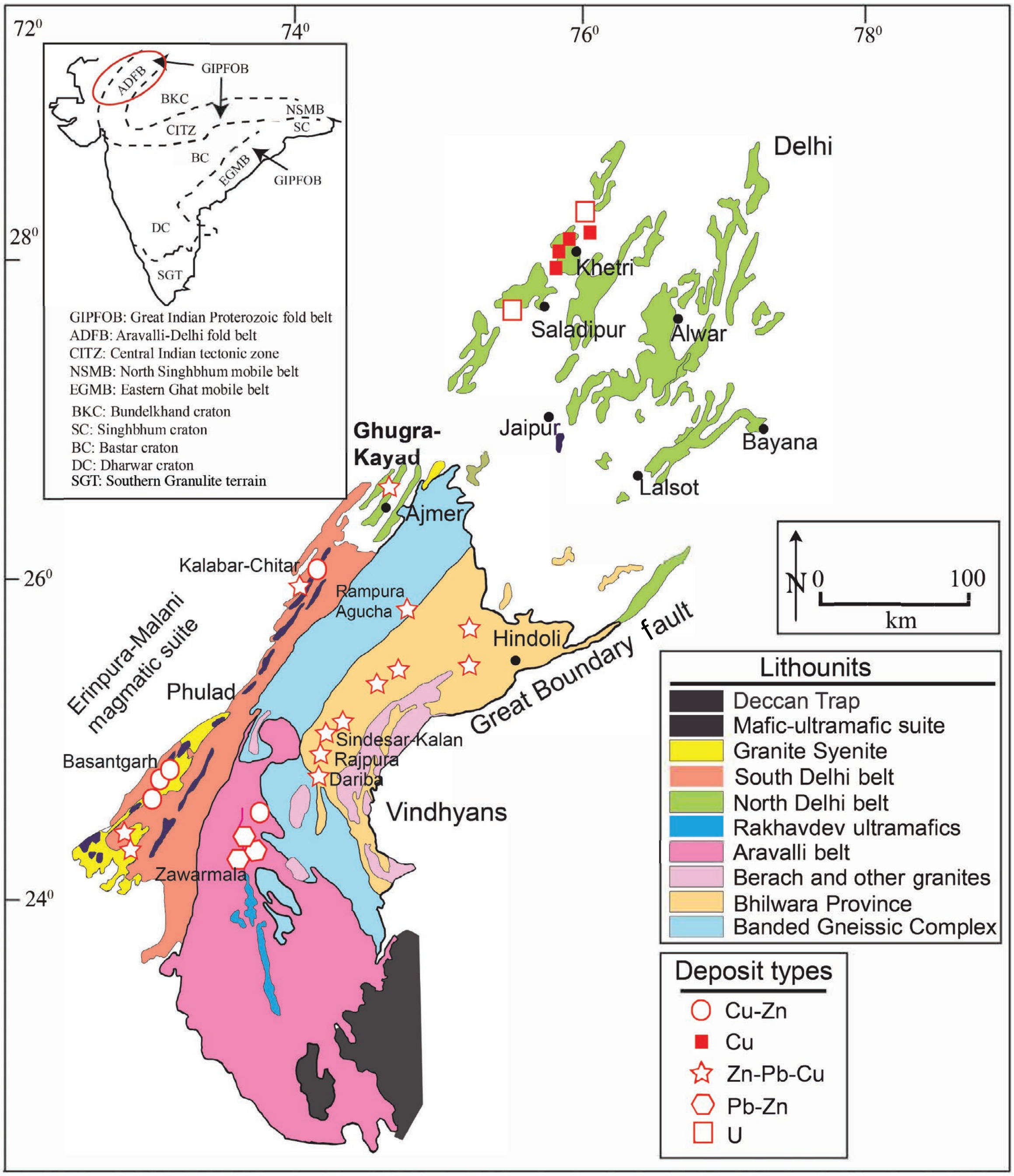

Fig. 1. Geologic map of the Delhi Aravalli fold belt with the lithological units and locations of different deposit types (redrawn from Deb et al., 2001).

Geologic Background

The Aravalli-Delhi fold belt, along with the Central Indian tectonic zone, the Eastern Ghats mobile belt, and the Southern Granulite terrain, is considered to be a part of the Great Indian Proterozoic fold belt (Radhakrishna and Naqvi, 1986; Leelanandam et al., 2006; Fig. 1 inset). The NE-SW–trending Aravalli-Delhi fold belt (Fig. 1) lies unconformably over the Archean Banded Gneissic Complex and comprises multiply deformed and metamorphosed conglomerate, quartzite, carbonaceous phyllite, impure marble, stromatolitic phosphate rock, and basic volcanic rocks (Ghosh et al., 2022 and references therein). The Mesoproterozoic to Neoproterozoic Delhi Supergroup, which forms a part of the linear Delhi fold belt, overlies the Paleoproterozoic Aravalli Supergroup and comprises carbonates and mafic volcanic rocks of the Raialo Group, arenaceous rocks and mafic volcanic rocks of the Alwar Group, and calcareous and argillaceous rocks with fewer volcanic rocks of the Ajabgarh Group (Roy and Jakhar, 2002; Sarkar and Gupta, 2012 and references therein). The Delhi fold belt is divided into the North Delhi fold belt and the South Delhi fold belt, Ajmer being the geographical dividing point. The North Delhi fold belt is characterized by horst-graben structures in the pre-Delhi basement rocks (Banerjee and Fareeduddin, 2020). Based on stratigraphy and metamorphic evolution, the North Delhi fold belt is divided into three subbasins, namely Khetri, Alwar, and Bayana-Lalsot (Fig. 1). The metapelites of the Khetri basin record two regional metamorphic events of which peak temperature-pressure conditions of 550° to 600°C and 5.5 kbar were attained during M2 metamorphism (Sarkar and Dasgupta, 1980). The Alwar subbasin has undergone two prograde metamorphic phases where M2 records peak conditions of 645°C at 7 kbar (Pant et al., 2008; Naik et al., 2022), whereas the Bayana rocks are less deformed than Alwar and attained lower greenschist facies conditions (Sarkar and Gupta, 2012; Mehdi et al., 2015). The present study area is located near Ajmer in the southernmost sector of the North Delhi fold belt (Fig. 1).

The Kayad deposit (26°31′50.37″N, 74°41′24.96″E) is located 9 km north-northeast of Ajmer city. Most of the lithounits in the area are sporadically exposed due to thick soil cover (Fig. 2A). Much of what is known about the geology of the Kayad deposit is from the initial explorations by the Geological Survey of India and premining surveys conducted by Hindustan Zinc Ltd., which is currently mining the deposit. The Kayad-Ghugra belt forms a part of the east-central Proterozoic belt of the Delhi Supergroup. The rocks of the Ajmer-Sambhar area are represented by the upper Ajmer Formation of the Ajabgarh Group, which is underlain by the Anasagar migmatites. The migmatites include biotite-rich paragneisses, amphibolites, metapsammites, xenolithic biotite-hornblende orthogneisses, and granite intrusives (Fareeduddin et al., 2014). Mukhopadhyay et al. (2000) reported an age of 1849 ± 8 Ma for the granite gneisses based on single-zircon evaporation method. Anasagar granite, which is intrusive into the gneisses, has been dated at 1600 ± 90 Ma by Choudhary et al. (1984). The metasedimentary package of the Ajmer Formation is represented by quartzite, carbonate, and thick sequences of metamorphosed arkose-pelite-graywacke (metapsammopelite). These rocks have undergone upper amphibolite facies metamorphism (garnet-staurolite to garnet-sillimanite grade) synchronous with the second of four phases of regional deformation (Fareeduddin et al., 2014). Thermobarometric studies conducted on syn- to post-D_2 garnet and staurolite porphyroblasts in metapelites of the Ajmer-Puskar section yielded peak P-T estimates of 5.7

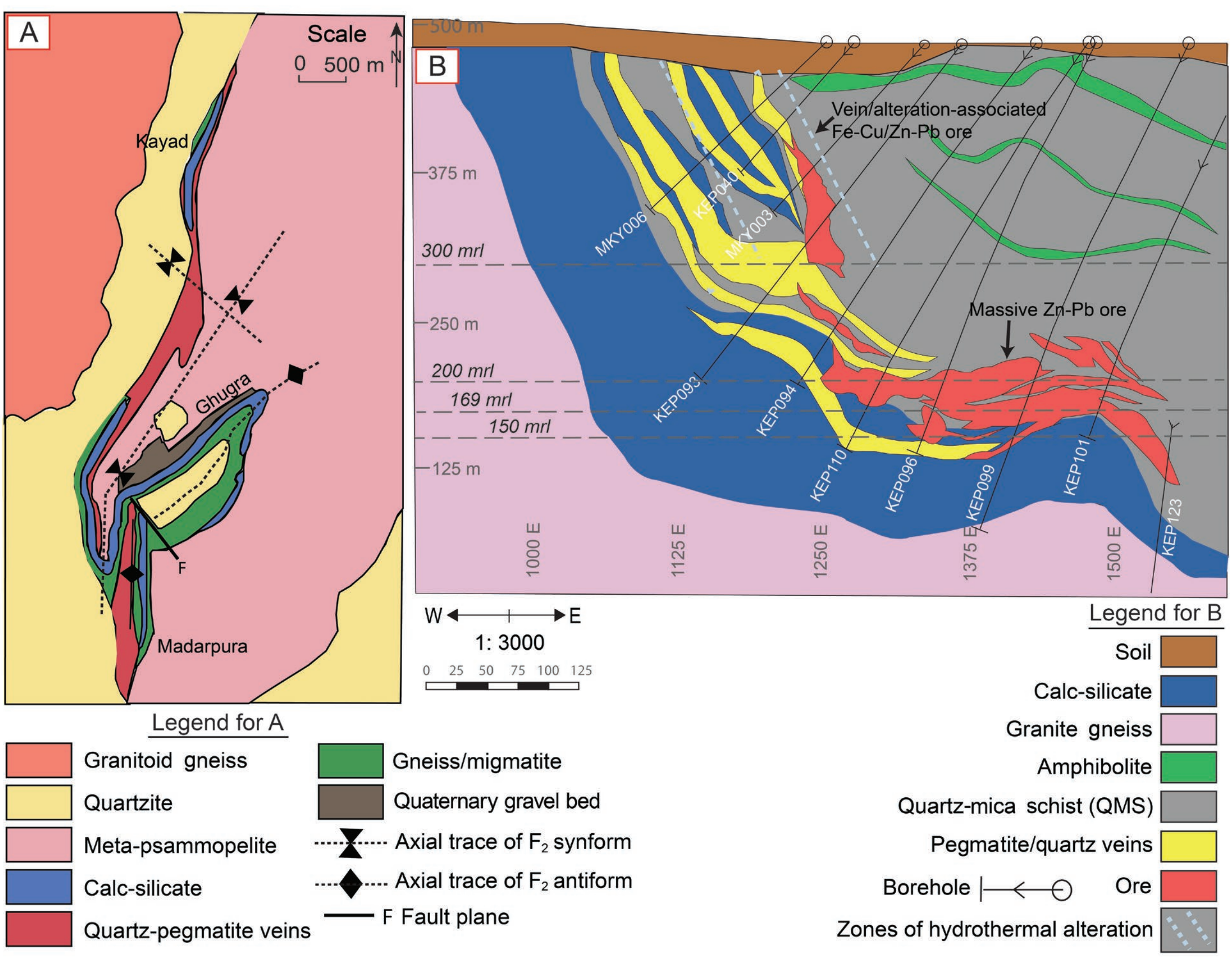

Fig. 2. A) Geologic map of Kayad (taken from GSI report by Chattopadhyaya, 1992); B) Schematic cross-sectional map of Kayad deposit, marked with the sample locations (courtesy Hindustan Zinc Ltd.). The host lithounits occur as a synformal fold with the ores at the hinges and eastern limb. Note the localization of massive Zn-Pb ores and vein/alteration associated Fe-Cu/Zn-Pb ores at the hinge and footwall, respectively. Horizontal dashed lines mark the depths at which samples for the study were collected. Abbreviation: mrl = mean reduced level.

± 1.5 kbar and 560° ± 50°C (Chattopadhyay et al., 2012), in agreement with the regional amphibolite facies metamorphic conditions.

The Kayad base metal deposit forms a part of the 20-km-long and 4-km-wide Ajmer Zn-Pb belt (Fareeduddin et al., 2014; Fig. 2A). The general trend of the formations in Kayad is north-northeast–south-southwest to north-east, coinciding with the regional trend of the axis of the Delhi synclinorium. The transverse section of the Kayad deposit (Fig. 2B) shows an open synformal fold structure with a shallow NE–plunging hinge that strikes N25°E-S25°W. The western limb dips about 60° to 70° due southeast, whereas the eastern limb has shallower dips ranging between 10° and 40° due southeast. Primary bedding is preserved in the form of compositional banding as well as alternating fine- and coarse-grained layering in metasedimentary rocks. The schistosity is parallel to the axial plane of the F_2 folds and constitutes the dominant tectonic fabric in the rocks.

The rock types exposed at different mine levels include calc-silicate rocks, pegmatite, quartz-mica-schist, and amphibolite (Figs. 2B, 3). Massive high-grade ores occur mostly in graphite-bearing quartz-mica schist with total reserves of 9.2 million tons at an average grade of 9.3% Zn and 1.2% Pb. Sphalerite and galena constitute the major ore minerals, together with minor to trace amounts of chalcopyrite, pyrrhotite, arsenopyrite, and pyrite. The main ore lens of the deposit varies in width from 5 m in the steeper portions to 35 m in the flatter hinge areas. The massive ore is mainly localized along the fold hinges (Fig. 2A). A zone of pegmatite/quartz veins with evidence of brecciation and silicification accompanied by profuse hydrothermal alteration is seen on the footwall side of the massive ore close to the contact between calc-silicate rocks and quartz-mica schist (Fig. 2B). Abundant chalcopyrite and pyrrhotite with lesser sphalerite and galena are closely associated with or hosted by these veins.

Samples and Methods

Samples used in this study were collected from various mining levels (125, 150, 169, 200, and 300 mean reduced level [mrl, in meters]; mine surface level = 487 mrl) of the Kayad underground mine (Fig. 3). Additional samples were collected from the mine dump and stockpile, where the approximate locations of the samples in the mines are known (Table 1).

Petrography and major and trace element analysis

Petrographic studies were carried out on 40 representative polished thin sections selected from all rock types hosting mineralization using an optical microscope (NIKON-100POL), a JEOL JSM-6490 scanning electron microscope (SEM) with an Oxford X-MaxN silicon drift energy dispersive spectrometry (EDS) detector, and a ZEISS EVO 18 Analytical SEM. Backscattered electron (BSE) images were used for characterizing textural relationships and intragrain compositional heterogeneities. The SEM analyses were done at the Department of Geological Sciences, Jadavpur University, and at the Department of Geology and Geophysics, Indian Institute of Technology (IIT), Kharagpur. The major element composition of the sulfide phases was determined on a Cameca SX-100 electron probe microanalyzer (EPMA) equipped with four wavelength dispersive spectrometers at the Department of Geology and Geophysics, IIT, Kharagpur. The analyses were conducted at a 20-kV accelerating voltage and 20-nA beam current with a ca. 1-µm beam diameter. A total of 99 sphalerite, 75 galena, 48 chalcopyrite, 72 pyrrhotite, and 36 arsenopyrite grains, across all types of mineralization, were analyzed for their major and minor element concentration. The X-ray lines used for the analysis of S, Fe, Cu, Zn, Pb, and As are SKα, FeKα, CuKα, ZnKα, PbMα, and AsLα respectively. The peak and background acquisition times were set at 10 s and 5 s, respectively, for all elements except As, for which the peak and background times were 20 s and 10 s, respectively. The standards used for calibration of the instrument are as follows: sphalerite for Zn, pyrite for Fe and S, galena for Pb, pure copper metal for Cu, and synthetic GaAs for As. The major and minor element compositions of the silicate phases were measured on a CAMECA SXFive EPMA at Mantle Petrology Lab, DST-SERB National Facility, Department of Geology (Center of Advanced Study), Insititute of Science, Banaras Hindu University. The operating conditions were 15-kV accelerating voltage and 10-nA beam current with an LaB6 electron source. The X-ray lines used in the analyses are as follows: F-Kα, Na-Kα, Mg-Kα, Al-Kα, Si-Kα, P-Kα, Cl-Kα, K-Kα, Ca-Kα, Ti-Kα, Cr-Kα, Mn-Kα, Fe-Kα, and Ba-Lα. Fluorite, albite, halite, periclase, peridot, corundum, wollastonite, apatite, orthoclase, rutile, chromite, rhodonite, hematite, and barite were used as natural mineral standards; pure metal was used as a standard for Ni.

The concentration of trace elements in the sulfide phases was measured on a Thermo Fisher Scientific iCap-Q quadrupole-inductively coupled plasma-mass spectrometer (Q-ICP-MS) coupled with a New Wave Research 193-nm ArF excimer laser ablation system at the Department of Geology and Geophysics, IIT, Kharagpur. The laser was operated at a 5-Hz repetition rate and ca. 5-J/cm^2 fluence at spot sizes of 45 to 50 µm. The instrument was optimized for maximum sensitivity using the United States Geological Survey (USGS) sulfide standard MASS-1. The raw counts for each isotope were acquired in time-resolved mode with 30 s of gas blank measurement and 45 s of peak signal measurement. Instrumental mass bias and drift were corrected by external standardization, with ten measurements of unknowns (samples) bracketed by two measurements of the MASS-1 sulfide standard. The data was reduced offline using the GLITTER© data reduction software. The isotopes that were measured include ^{51}V, ^{55}Mn, ^{59}Co, ^{60}Ni, ^{63}Cu, ^{71}Ga, ^{73}Ge, ^{75}As, ^{82}Se, ^{95}Mo, ^{107}Ag, ^{111}Cd, ^{115}In, ^{118}Sn, ^{121}Sb, ^{193}Ir, ^{195}Pt, ^{197}Au, ^{202}Hg, ^{205}Tl, and ^{208}Pb. A total of 62 spots on sphalerite, 43 on pyrrhotite, 40 on galena, and 26 on chalcopyrite were measured. Care was taken to select clean, inclusion-free grains with radii >40 µm to ensure minimal contamination from inclusions or neighboring minerals. The concentrations of V, Ir, Au, and Pt are consistently below detection limits in all the sulfides.

Thermometry

The metamorphic conditions of the ore and host rocks of the Kayad deposit were estimated using several thermometric formulations. The trace element composition of sphalerite is sensitive to its temperature of formation/reequilibration. Sphalerite formed at high T can accommodate high propor-

Table 1. Summary of Mineralization Style, Host Rocks, Vein Mineralogy, Ore Mineralogy, and Mineral Alteration Features and Hydrothermal Assemblages

Sulfide mineralization category	Rock type	Host-rock/vein mineralogy	Sulfide mineralogy	Associated alteration features
Disseminated/ laminated	Quartz-mica schist	Quartz, muscovite, biotite, oligoclase-andesine, graphite, tourmaline, apatite	Sphalerite, pyrrhotite ± chalcopyrite ± galena	No alteration
	Quartzite	Quartz, tourmaline, graphite ± feldspar ± muscovite ± biotite	Sphalerite, pyrrhotite, chalcopyrite	No alteration; chamosite and albite in vicinity of pyrrhotite veins intruding in quartzite host
Massive	Quartz-mica schist	Quartz, muscovite, biotite, oligoclase-andesine, graphite, tourmaline, apatite	Sphalerite, galena, arsenopyrite, pyrrhotite, chalcopyrite, ± sulfosalts	No alteration in and around the massive ores; replacement of oligoclase-andesine by K-feldspar and albite; assemblage of monazite + allanite-clinozoisite + apatite in the vicinity of pyrrhotite-chalcopyrite–rich massive Zn-Pb sulfide
Vein/veinlets	Pegmatite	Vein mineralogy: oligoclase, quartz, muscovite	Pyrrhotite, chalcopyrite ± sphalerite ± galena	Pegmatitic oligoclase-andesine is replaced by K-feldspar + albite; pegmatitic muscovite is replaced by K-feldspar +albite+ biotite + chamosite; alteration assemblage of Al-pumpellyite + K-feldspar + albite + graphite; increased proportion of coarse-grained biotite and patchy graphite in the vein wall
	Veinlets in calc-silicate	Vein mineralogy: pyrrhotite, sphalerite ± chalcopyrite, sphene, calcite, hornblende	Pyrrhotite, sphalerite ± chalcopyrite	Sphene, chlorite, epidote, calcite, hornblende
	Microcline vein-hosted	Quartz mica schist mineralogy: muscovite, biotite, quartz, plagioclase > orthoclase, tourmaline, apatite, graphite Vein mineralogy: microcline, quartz ± biotite, ± tourmaline	Spahelrite, galena, pyrrhotite, ± chalcopyrite ± arsenopyrite	Prehnite + pumpellyite + clinochlore ± albite replacing K-feldspar; assemblage of monazite + allanite-clinozoisite + apatite in the vicinity of mineralization

tions of Fe, Mn, In, and Sn, whereas that formed at low T typically has elevated concentrations of Ga, Ge, As, Sb, Tl, and Ag, which may differ by two to three orders of magnitude (Stoiber, 1940; Cook et al., 2009; Frenzel et al., 2016; Bauer et al., 2019; Li et al., 2020). We used the thermometer developed by Frenzel et al. (2016) (using concentrations of Ga, Ge, In, Mn, and Fe and defined as the GGIMF thermometer by the authors) to estimate the temperature based on 62 sphalerite analyses. The As concentration of arsenopyrite changes with the mineralogical association and temperature. The minimum and maximum temperatures were calculated from arsenopyrite-löllingite association in massive ores using atomic % As in arsenopyrite in the T-X diagram of Kretschmar and Scott (1976). Biotite formula was calculated using the Excel spreadsheet of Li et al. (2022) and temperature was estimated using the Ti-in-biotite thermometer of Wu and Chen (2015) (revised after Henry et al., 2005), which is applicable over a temperature and pressure range of 480° to 840°C and 0.1 to 1.9 GP and at XTi values of 0.02 to 0.14. The prerequisite for the application of this thermometer is the presence of Ti-bearing minerals such as ilmenite or rutile, of which the latter is found in abundance in the rocks at Kayad.

Results

Rock types

Quartz-mica schists (QMS) are fine- to medium-grained, schistose, and composed predominantly of quartz, feldspar, muscovite, biotite, and graphite (Fig. 4A-B). Minor minerals include tourmaline, rutile, titanite, and apatite with infrequent ilmenite and zircon. Quartz and feldspar are subhedral; some grains are elongated in the direction of the pervasive schistosity. Oligoclase-andesine and orthoclase feldspars far exceed albite proportions. Tourmaline (50–300 μm) is subrounded to elongate. Rutile and titanite occur as fine (up to 30–40 μm) subhedral to anhedral grains whereas apatite grains are subrounded and of equivalent size to rutile and titanite. The rocks are highly deformed and folded at different scales. The quartz-mica schist hosts the bulk of the Zn-Pb ores.

Quartzites are compact, fine to medium grained, and composed predominantly of quartz and variable amounts of oligoclase-andesine and muscovite>biotite. Titanite, rutile, tourmaline, and zircon are dispersed in the rock as accessory phases. At places, the rocks have a banded appearance with alternating graphite + mica-rich bands (Fig. 4C). The mineralogy of these graphite + mica-rich bands is similar to the quartz-mica schists, except for the higher proportion of quartz. Minor sulfide mineralization is hosted by this rock.

Calc-silicate rocks form the lower bounding strata or the footwall margin of the main orebody in Kayad (Fig. 2B). They are medium to coarse grained, ferruginous, and composed predominantly of diopside, augite, and tremolite, with minor quartz and apatite (Fig. 4D). The calcsilicates contain minor sulfide mineralization in the form of veinlets.

Pegmatites of two different generations are seen in the mine. The older pegmatites are folded along with the strata, whereas the younger ones mainly occur at the contacts of calcsilicates and quartz-mica schist in the footwall of the

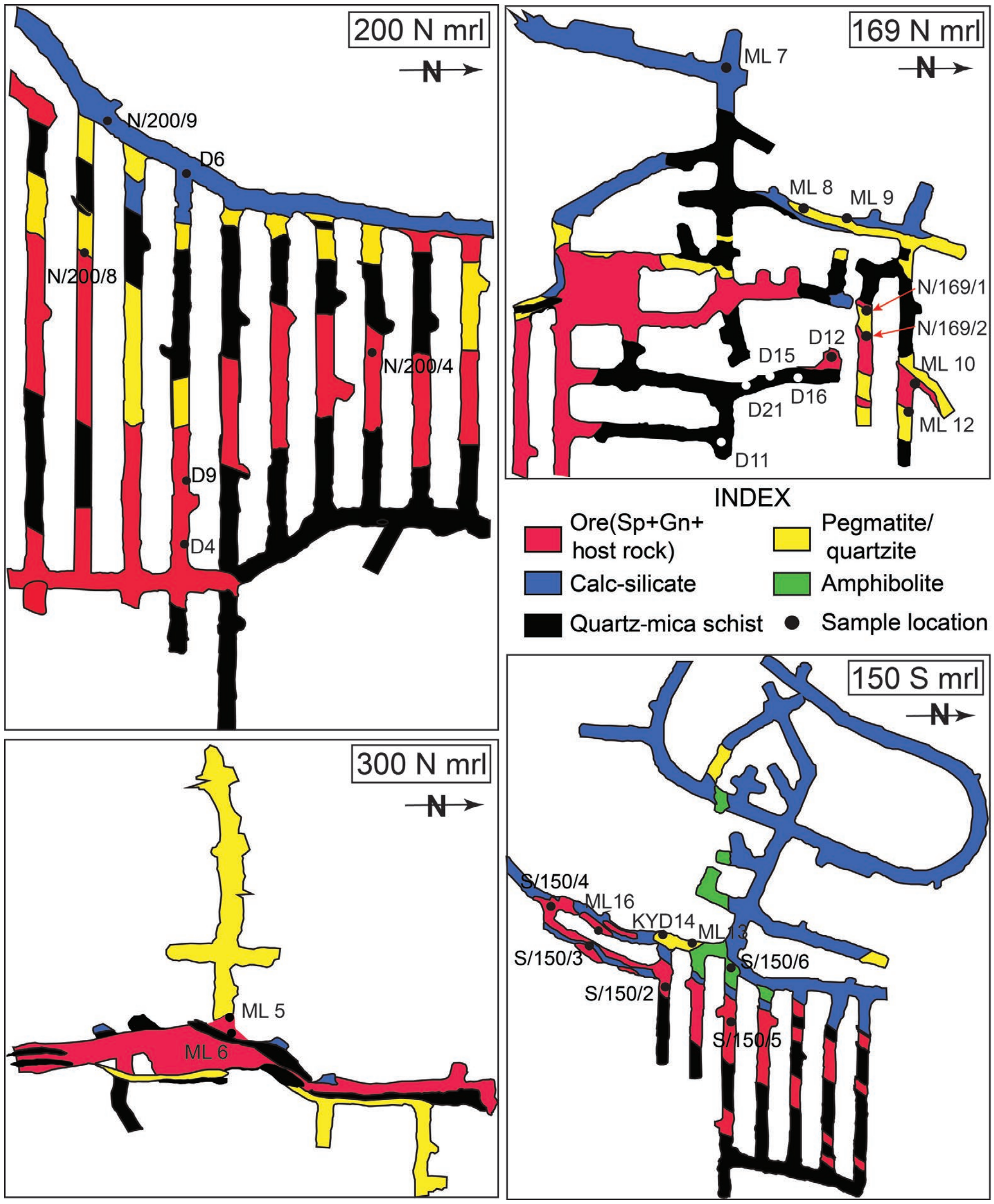

Fig. 3. Plan maps of levels from which sampling was performed with sample locations (courtesy Hindustan Zinc Ltd.). Alphanumeric values on the maps are sample numbers. Mrl = mean reduced level. Abbreviations: Gn = galena, mrl = mean reduced level, Sp = sphalerite.

massive Zn-Pb orebody (Fig. 2B). They are composed of quartz, oligoclase, and microcline with occasional tourmaline and books of muscovite. Quartz veins are spatially associated with the younger pegmatites. Sulfide mineralization occurs in both the younger pegmatites (Fig. 4E-F) and the quartz veins (Fig. 4G).

Mode of occurrence of sulfides and associated alteration features

Disseminated/laminated sphalerite-pyrrhotite ± chalcopyrite ± galena: Disseminated and laminated sphalerite-pyrrhotite ± chalcopyrite ± galena occur in graphite-bearing quartz-mica schist and quartzite. These ores are rarely noticeable in the underground mine since the mine was developed primarily to extract the high-grade massive ores. However, microscopic studies based on borehole and limited mine samples show sphalerite and pyrrhotite with minor amounts of chalcopyrite and galena (in decreasing order of abundance) aligned parallel to the foliation defined by mica and graphite in the schists (Fig. 5A). Sphalerite typically contains numerous chalcopyrite inclusions, a texture commonly referred to as chalcopyrite disease. In general, the laminae containing the ore minerals are cofolded with the schistosity of the host rock. Stumpy aggregates of sphalerite, galena, and euhedral to subhedral arsenopyrite occur at the hinges of microfolds. No visible alteration is seen in association with these sulfides. However, the rock at places contains clusters of chalcopyrite and pyrrhotite disrupting the foliation. In the vicinity of these Fe-Cu sulfides,

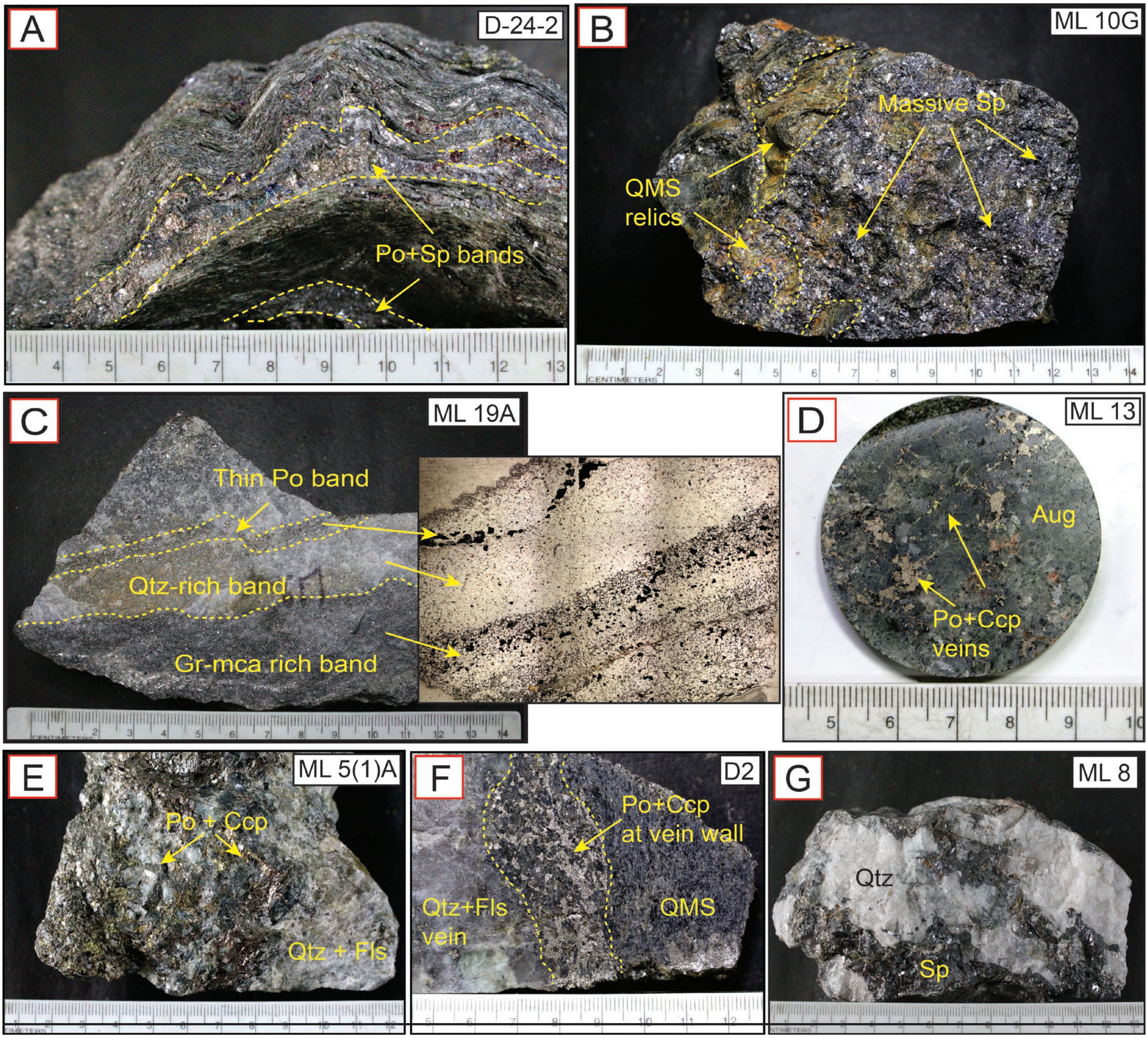

Fig. 4. Hand specimen photographs of different host rocks. A) Laminated quartz-mica schist (QMS) with bands/laminae of sulfides. B) Massive Zn-Pb ores with relics of host quartz-mica schist. C) Banded quartzite with alternate graphite-mica–rich and quartz-rich bands; the transmitted light photomicrograph is shown on the right (at the arrow-head); note the enrichment of sulfide (black) in the graphite-rich band. D) Calcsilicate with Fe-Cu sulfide veins. E) Pegmatite with abundant pyrrhotite and chalcopyrite mineralization. F) Pyrrhotite and chalcopyrite mineralization at pegmatite vein wall at the contact with quartz-mica schist. G) Quartz vein with sphalerite + galena ± pyrrhotite ± chalcopyrite. Abbreviations: Aug = augite, Ccp = chalcopyrite, Fls = feldspar, Gr-Mca = graphite-mica, Po = pyrrhotite, Qtz = quartz, Sp = sphalerite. Alphanumeric at the top right of each subfigure represents sample number.

oligoclase-andesine of the host rock is replaced by either orthoclase and albite (Fig. 6A) or by Ce-allanite + Ce-monazite + fluorapatite that occur in clusters along with Fe-Cu sulfides, typically with monazite cores rimmed by allanite and fluorapatite (Fig. 6B).

Quartzite contains limited mineralization, primarily as sphalerite and pyrrhotite disseminations with sparse chalcopyrite and galena (10–20 µm in size) in decreasing order of abundance. In the compositionally layered samples, sulfides cluster preferentially in the graphite-rich layers (Figs. 4C, 5B). Sphalerite in quartzite is relatively clean with minimal chalcopyrite disease. No visible alteration is seen around the disseminated sulfides. However, pyrrhotite veins that intrude and taper into the rocks have an alteration halo characterized by coarse quartz, Fe-rich clinochlore, and an increased proportion of biotite and K-feldspar in the host rock at the vein contact (Fig. 6C). The coarse K-feldspar grains are locally replaced by albite, Ce-monazite, Ce-allanite, fluorapatite, and at places clinozoisite, similar to the alteration halo in pyrrhotite-chalcopyrite–bearing assemblages that overprint the foliation in the quartz-mica schist (Fig. 6D).

Pyrite is sparse in these associations. However, rare relics of pyrite within pyrrhotite are seen at different places in the quartzite and quartz-mica schist (Fig. 5C).

Massive sphalerite-galena ± pyrrhotite ± chalcopyrite: Massive ore in graphite-bearing quartz-mica schist predominantly comprises sphalerite and galena with variable proportions of chalcopyrite, pyrrhotite, and arsenopyrite. The sulfides occur in large masses, disrupting the pervasive tectonic foliation of the rocks. In massive ores, sphalerite forms the matrix, with the other sulfide phases occurring as inclusions or irregular blebs that are a few tens to hundreds of microns in size (Fig. 5D). Galena exhibits cusp-and-carries texture with sphalerite (Fig. 5E). Fragments of contorted and folded quartz-mica schist occur in the massive sphalerite matrix, in a prominent "durchbewegung" texture (Marshall and Gilligan, 1989; Fig. 5F). Large euhedral grains of arsenopyrite, 500 to 2,000 µm across, occur within the massive sulfide matrix and also cut across the foliation in the vicinity of massive ores (Fig. 5D). Most of these arsenopyrite grains have relicts of löllingite with sharp contacts (Fig. 5G). A characteristic feature of the massive ore is the frequent occurrence of a variety of sulfosalt grains and aggregations. They are typical components of the Pb-Zn massive ores in other deposits of the Aravalli-Delhi fold belt, such as Zawar, Rampura-Agucha, Rajpur Dariba, and Sindesar-Khurd (Sarkar and Gupta, 2012 and references therein). Sulfosalts identified in the Kayad deposit are gudmundite (FeSbS), pyrargyrite (Ag_3SbS_3), breithauptite (NiSb), Ag tetrahedrite ($(Cu, Fe, Ag)_{12}(Sb, As)_4S_{13}$), and some unidentified phases having variable amounts of Cu, Fe, Ag, Sb, and Zn. In most cases, sulfosalts occur as numerous discrete phases or are closely intergrown with other sulfides. For example, gudmundite is commonly intergrown with pyrrhotite and galena with highly irregular and lobate grain boundaries (Fig. 5H). Similar textural relations are seen among pyrargyrite, gudmundite, galena, pyrrhotite, and chalcopyrite (Fig. 7A). Gudmundite also shares irregular to nearly straight grain boundaries with galena and sphalerite (App. Fig. A1). In places, the volume of gudmundite (approximated from its surface area) is nearly equal to that of the associated sulfides such as sphalerite (Fig. 7B). Breithauptite, in general, occurs as idiomorphic grains when in contact with galena (Fig. 7C; App. Fig. A1), and as irregular or somewhat elongate grains when juxtaposed against pyrrhotite or (rarely) sphalerite (Fig. 7D; App. Fig. A1). Silver-bearing tetrahedrite either shares straight grain boundaries with galena (Fig. 7E) or is present as inclusions in the latter (Fig. 7F; App. Fig. A1). Scant Sb-rich phases (Sb >80 wt %) and mixed compositions of Ag-Sb are observed in some places (App. Fig. A1).

Silicate phases associated with the massive sulfides are significantly coarser grained than those in the laminated quartz-mica schist. The sulfides are often encapsulated by a film of recrystallized quartz (Fig. 7G). In the foliated parts, quartz, mica (muscovite>biotite), Ca-plagioclase (mostly andesine), and orthoclase constitute the host; however, an increased percentage of recrystallized microcline is observed around the massive sulfides. Biotite and muscovite grains are randomly distributed around the coarser sulfides without any ordered alignment.

It is noted that in locales where chalcopyrite and pyrrhotite dominate over sparse sphalerite and galena in the massive ore, oligoclase and andesine of the host rock are altered along the cleavage planes to K-feldspar and fine specks of albite (Fig. 6E), similar to what is observed in the quartz-mica schist and quartzite (in section headed "Disseminated/laminated sphalerite-pyrrhotite ± chalcopyrite ± galena").

Vein-associated pyrrhotite-chalcopyrite ± sphalerite ± galena and sphalerite-galena ± pyrrhotite ± chalcopyrite: Sulfide mineralization associated with pegmatite/quartz veins and minor veinlets in the footwall of the massive orebody (Fig. 2B) is represented either by pyrrhotite-chalcopyrite association with or without sphalerite and galena (Fe-Cu ± Zn ± Pb) or by galena-sphalerite with or without pyrrhotite and chalcopyrite (Zn-Pb ± Fe ± Cu). Fe-Cu ± Zn ± Pb mineralization is present within the veins and/or in the vein walls (Figs. 4F, 7H). Within the veins, the sulfide assemblages are ubiquitously associated with alteration aureoles represented by the pervasive replacement of plagioclase by wart-like orthoclase and albite along grain boundaries and cleavage planes (Fig. 6F). At places, the alteration front comprising Fe-rich muscovite (muscovite in the alteration zone contains ~3 wt % FeO, whereas that in the pegmatite contains ~1.5 wt % FeO), biotite, chamosite, quartz, orthoclase, and albite replaces plagioclase (Fig. 6G) and muscovite (Fig. 6H). The secondary mineral pair of orthoclase and albite is ubiquitously accompanied by micropores. In zones of wall-rock mineralization, pyrrhotite and chalcopyrite are dispersed in the quartz-mica schist host, disrupting the schistosity (Fig. 4F). Biotite and graphite increase in abundance from the host rock toward the vein wall, gradually coarsening in size and being arbitrary in orientation. Coarse fluorapatite (20–100 µm) is abundant in the mineralized vein wall and is commonly rimmed by sulfides. Graphite is flaky in the host quartz-mica schist but is patchy to spherulitic in association with Fe-Cu sulfide (Fig. 8A).

Minor amounts of Fe-Cu ± Zn ± Pb mineralization consisting of pyrrhotite-chalcopyrite and sphalerite occur as thin veinlets in calc-silicates (Figs. 4D, 8B). The secondary minerals, represented by a combination of calcite, epidote, quartz, titanite, hornblende, albite, and chlorite (Fig. 8C), either occur in the veins or replace augite in the calc-silicates.

The Zn-Pb ± Fe ± Cu mineralization in veins is subordinate compared to the massive Zn-Pb ores in the fold hinges. However, they reach economic grade in some mining levels (Fig. 2B). Sphalerite grains in the quartz veins are coarse, exceeding a few centimeters in some instances (Fig. 4G). At places, the alteration assemblage is defined by specks of Al-pumpellyite, graphite, orthoclase, and albite (Fig. 8D). Microveins and clusters comprising K-feldspar (microcline > orthoclase), tourmaline, sphalerite, and galena are occasionally seen in the quartz-mica schist (Fig. 8E). Feldspar (up to 500 µm compared to 10–100 µm in the host rock) and tourmaline (up to 3 mm compared to 50 µm in the host) in these veins are much coarser than those in the host quartz-mica schist. Locally, microcline is further replaced by prehnite ± pumpellyite ± clino-

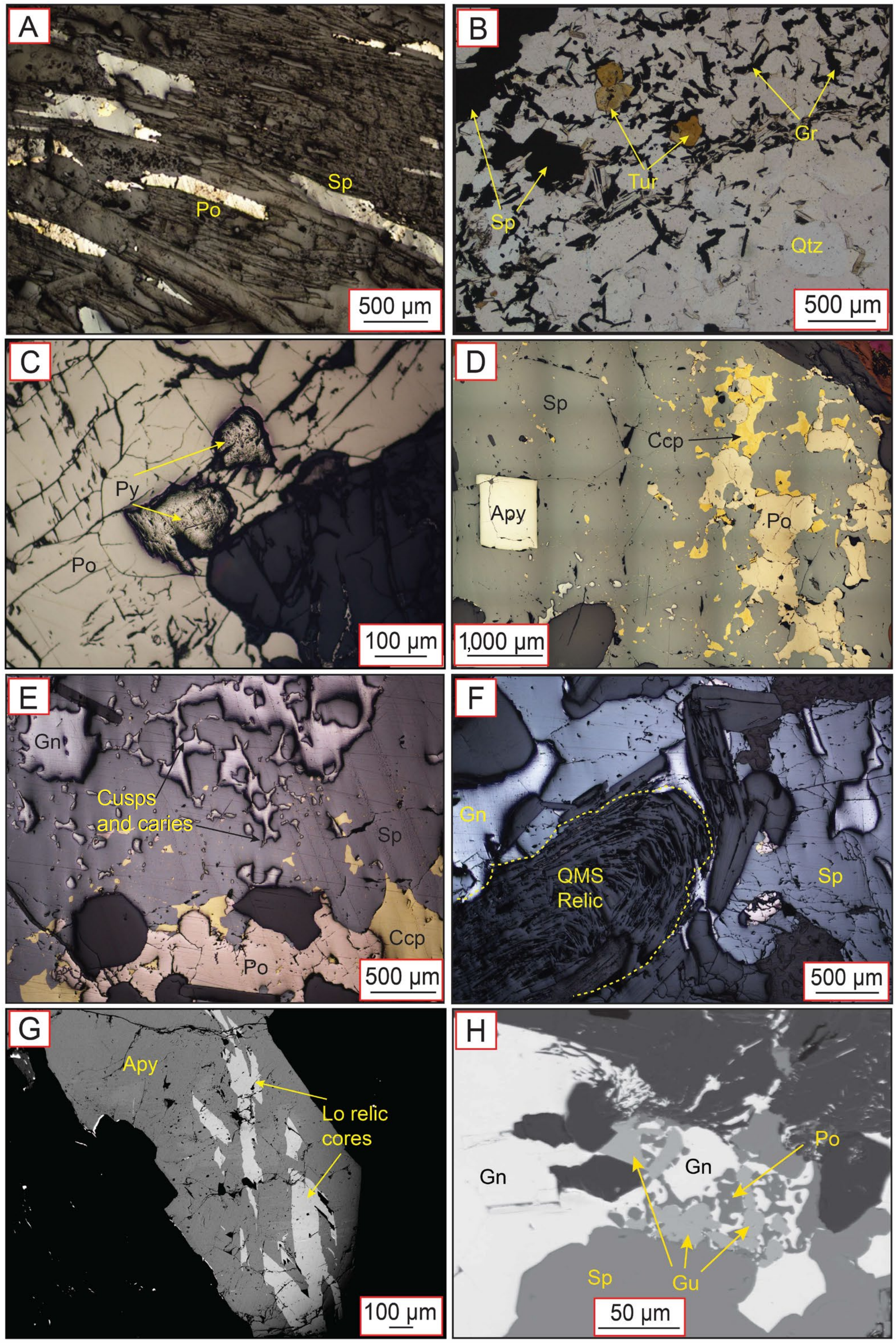

Fig. 5. Photomicrographs (A-F) and backscattered electron (BSE) images (G-H) showing the modes of occurrence and textures of sulfide mineralization. A) Laminated sphalerite and pyrrhotite parallel to general foliation in quartz-mica schist (QMS). B) Sphalerite-pyrrhotite and tourmaline grains in the graphite-rich band of compositionally banded quartzite. C) Irregular pyrite relics in pyrrhotite. D) Galena, pyrrhotite, chalcopyrite, and arsenopyrite in sphalerite matrix from massive Zn-Pb ores. E) Cusps and caries texture exhibited by massive sphalerite and galena. F) Folded quartz-mica schist relics in massive sulfides (Durchbewegung texture). G) Subhedral massive arsenopyrite grain with relic cores of löllingite. H) Intergrowth texture involving gudmundite, galena, and pyrrhotite in Zn-Pb massive ore. Abbreviations: Apy = arsenopyrite, Bt = biotite, Ccp = chalcopyrite Gn = galena, Gr = graphite, Gu = gudmundite, Lo = Löllingite, Po = pyrrhotite, Py = pyrite, Pyg = pyrargyrite, Qtz = quartz, Sp = sphalerite, Tur = tourmaline.

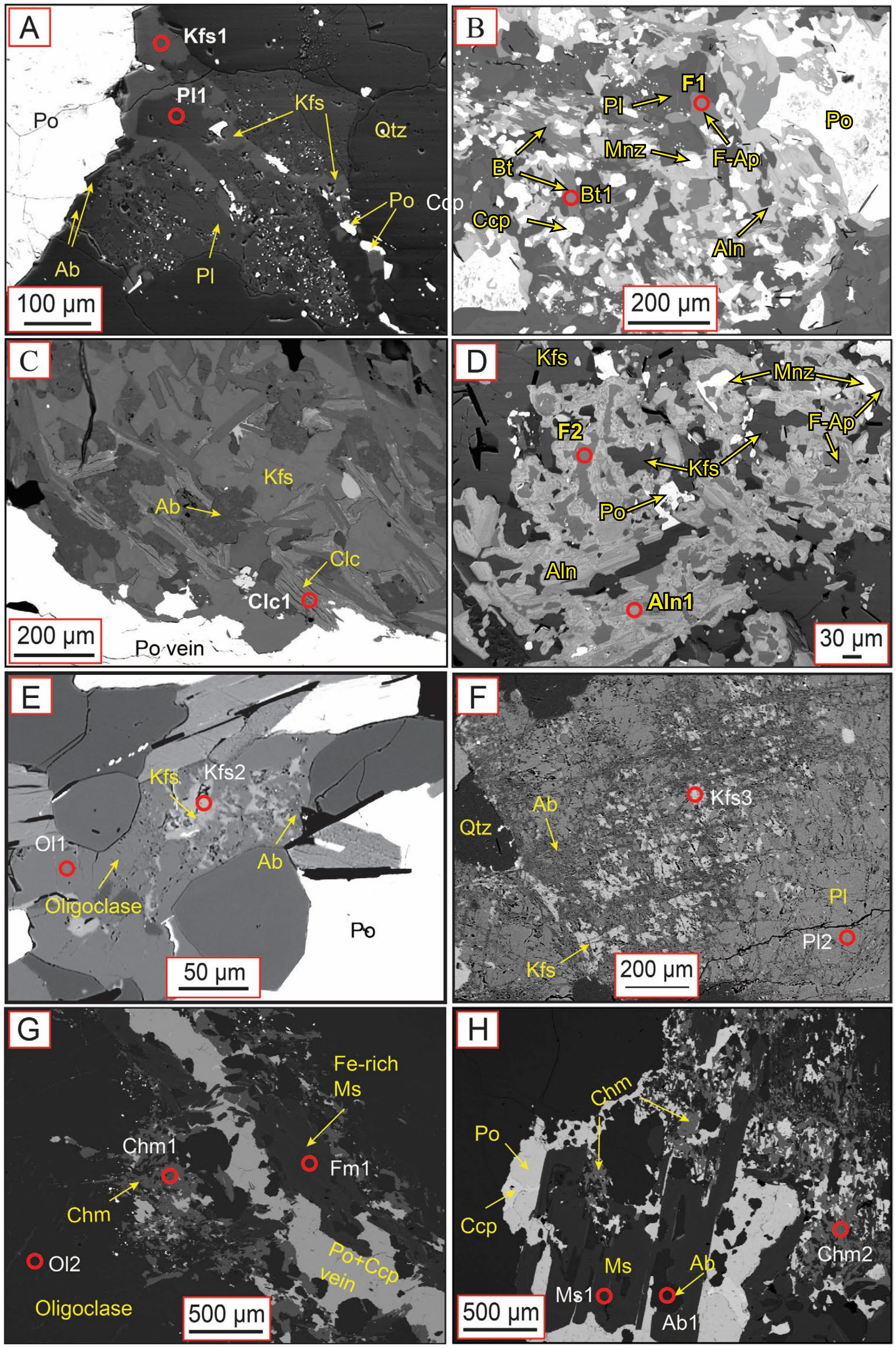

Fig. 6. Backscattered electron (BSE) images depicting alteration features associated with sulfide mineralization. A) K-feldspar and albite replace oligoclase in areas of intense Fe-Cu sulfide mineralization in quartz-mica schist. B) Monazite, allanite, and fluorapatite alteration assemblage coexists with pyrrhotite-chalcopyrite mineralization in quartz-mica schist, laminated type. C) Presence of clinochlore and replacement of K-feldspar by albite at the contact of the quartzite host and the pyrrhotite vein. D) Assemblage of monazite, allanite, fluorapatite, and K-feldspar with pyrrhotite in quartzite host. E) Orthoclase and cryptograins of albite replace oligoclase in Fe-Cu–rich areas in the massive ore. F) Within pegmatite veins, albite and K-feldspar pervasively replace oligoclase along its cleavage planes. G) Alteration halo around Fe-Cu sulfide vein comprising chamosite, Fe-rich muscovite, K-feldspar, and albite. H) Pegmatitic muscovite is replaced by chamosite, sulfides (pyrrhotite-chalcopyrite), Fe-rich muscovite, and albite. Alphanumeric values on figures refer to electron probe microanalysis (EPMA) spots provided in the Appendix table. Abbreviations: Ab = albite, Aln = allanite, Bt = biotite, Ccp = chalcopyrite, Chm = chamosite, Clc = clinochlore, F-Ap = fluorapatite, Fm = Fe-rich muscovite, Kfs = K-feldspar, Mnz = monazite, Ms = muscovite, Ol = oligoclase, Pl = plagioclase, Po = pyrrhotite, Qtz = quartz.

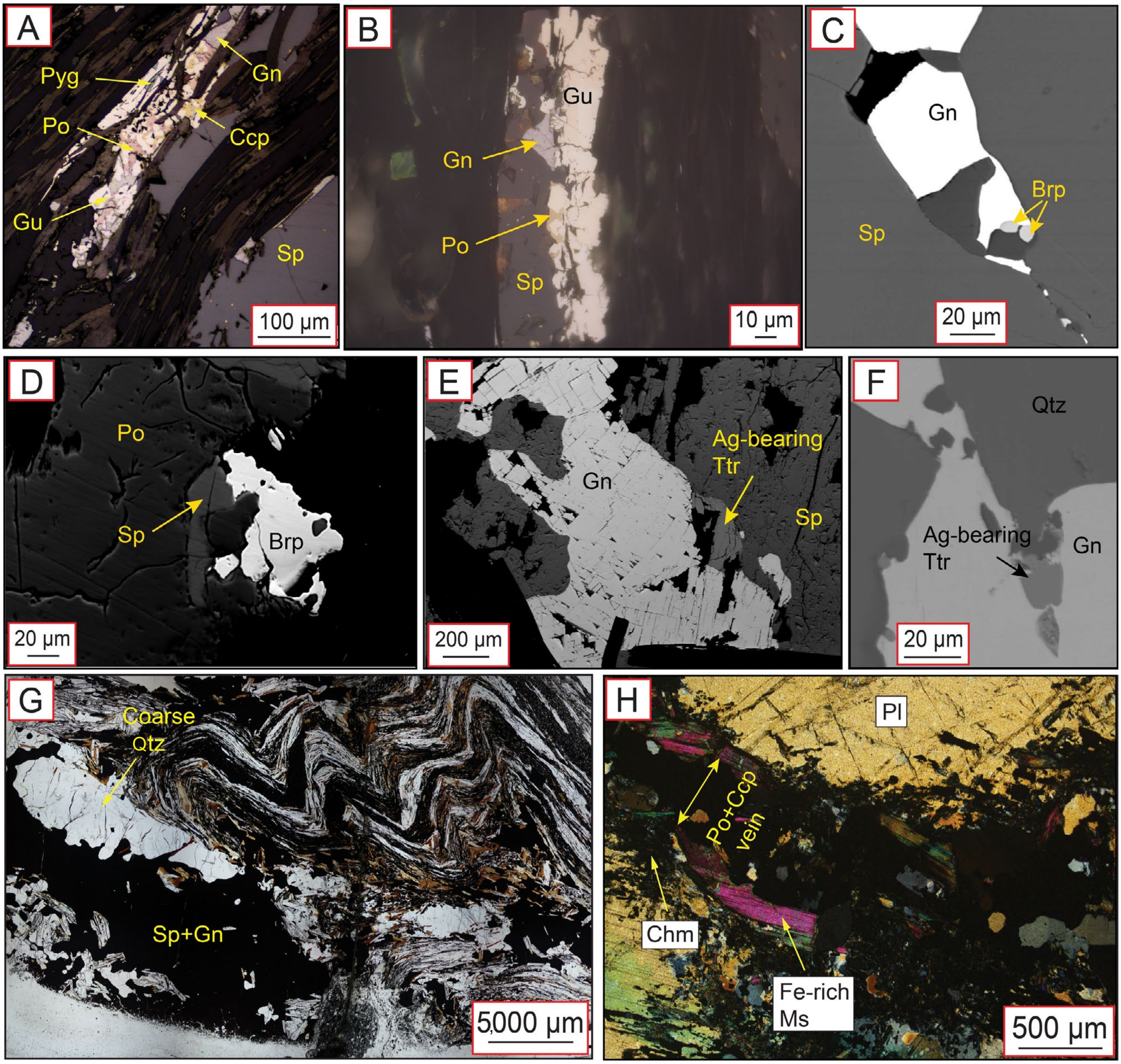

Fig. 7. Photomicrographs (A-B, G-H) and backscattered electron (BSE) images (C-F) of sulfosalts and textures of vein-hosted mineralization. A) Sulfosalts like pyrargyrite and gudmundite in association with galena, pyrrhotite, and chalcopyrite in quartz-mica schist-hosted massive Zn-Pb ores. B) Gudmundite association with sphalerite, galena, and trace pyrrhotite. C) Euhedral breithauptite crystals sharing grain boundaries with galena. D) Irregular breithauptite grain with pyrrhotite and minor sphalerite. E) Silver-bearing tetrahedrite grain with galena in quartz-mica schist. F) Intergrowth of Ag-bearing tetrahedrite and galena. G) Sphalerite and galena vein intruding along the limb of the microscopic fold, where sulfides also accumulate at the fold hinges. H) Pyrrhotite and chalcopyrite in pegmatite surrounded by secondary alteration minerals. Abbreviations: Brp = breithauptite, Ccp = chalcopyrite, Chm = chamosite, Gn = galena, Gu = gudmundite, Ms = muscovite, Pl = plagioclase, Po = pyrrhotite, Pyg = pyrargyrite, Qtz = quartz, Sp = sphalerite, Ttr = tetrahedrite.

chlore ± fluorite. Sphalerite and galena are scattered within the confines of this alteration mix (Fig. 8F). The alteration front comprising albite, sphalerite, and galena replaces K-feldspar at places, and the latter has a turbid appearance with patches of Na-rich domains (Fig. 8H). In places, monazite, fluorapatite, and allanite are associated with the sulfides. Ce-monazite is rimmed by a double corona of fluorapatite and a broad collar of allanite (Fig. 8G).

Major and trace element geochemistry

Sphalerite: Sphalerite contains around 8.0 to 12.6 wt % Fe with small inter- and intrasample variation. The trace element concentrations of all analyzed sphalerite grains (n = 62) are graphically presented in Figure 9A. Statistical parameters computed for the whole data set are provided as an electronic supplement (App. Table A1). We have checked the time-resolved downhole profile obtained during ablation for each

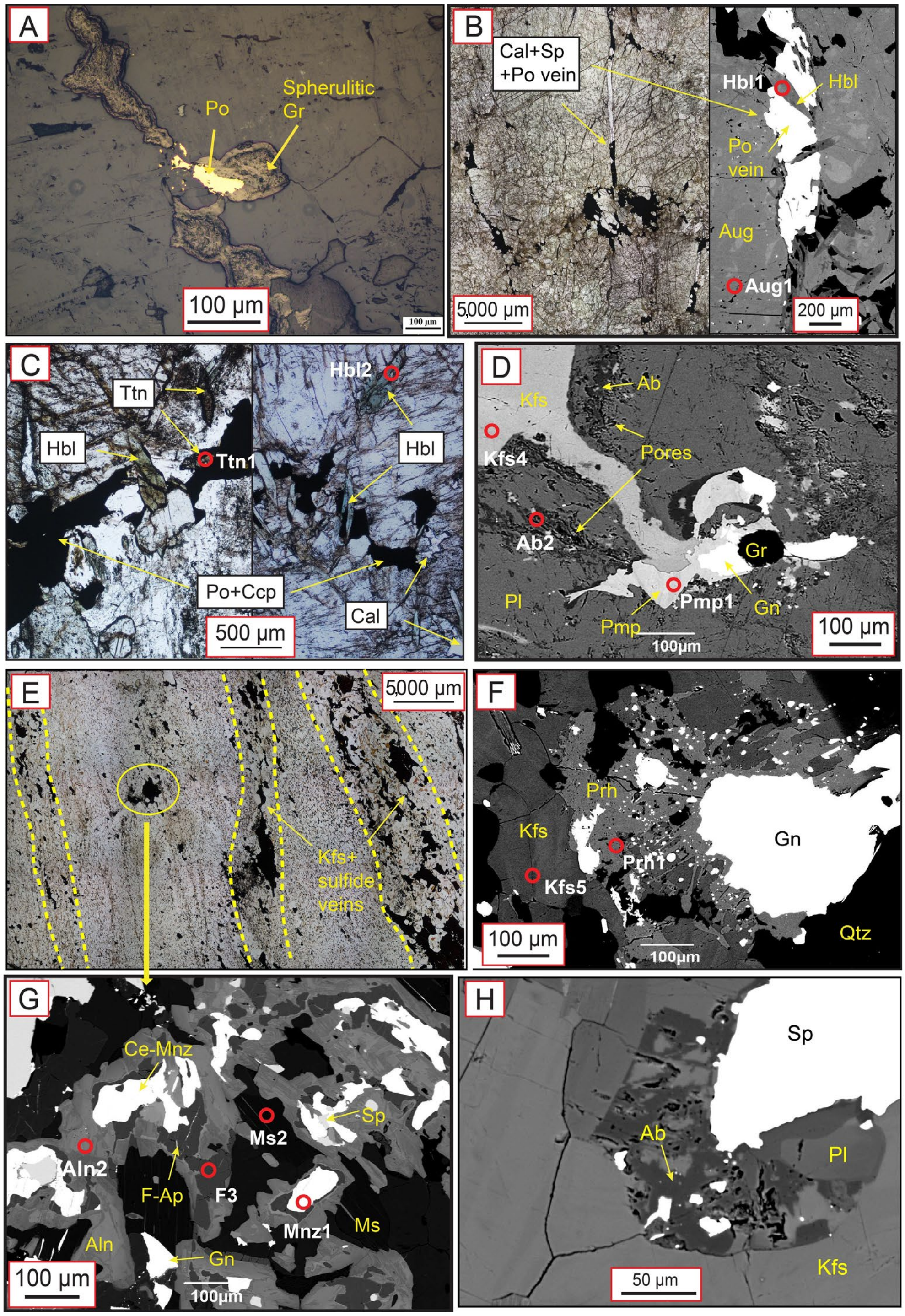

Fig. 8. Photomicrograph (A, B [right], C, E) showing textures in vein-hosted mineralization and backscattered electron (BSE) images (B [left], D, F-H) showing alteration patterns. A) Spherulitic variety of graphite associated with chalcopyrite and pyrrhotite veins in pegmatite. B) Sulfide (pyrrhotite-chalcopyrite-sphalerite) veinlets intrude into augite-rich calcsilicate rock. C) Veinlets are composed of hornblende, titanite, and calcite, which are ubiquitously seen in association with Fe-Cu sulfides in calcsilicate. D) Pumpellyite, spherical graphite, galena, and K-feldspar replace oligoclase within pegmatite vein. Innumerable pores (black pit-like appearance) have formed at sites of plagioclase replacement by albite. E) Fine microcline + orthoclase-bearing veins in quartz-mica schist rocks typified by sphalerite + galena mineralization (areas within in dotted lines). F) Prehnite envelops around galena and replaces K-feldspar of the same vein from (E). G) Allanite assemblage with Ce-monazite, fluorapatite, galena, and sphalerite in the above-mentioned veins of (E). H) Dark albite patches K-feldspar grains in the proximity of sphalerite-galena mineralization. Abbreviations: Ab = albite, Aln = allanite, Aug = augite, Cal = calcite, Ccp = chalcopyrite, Ce-Mnz = Ce-monazite, F-Ap = fluorapatite, Gn = galena, Gr = graphite, Hbl = hornblende, Kfs = K-feldspar, Mnz = monazite, Ms = muscovite, Pmp = pumpellyite, Pl = plagioclase, Po = pyrrhotite, Prh = prehnite, Qtz = quartz, Sp = sphalerite, Ttn = titanite.

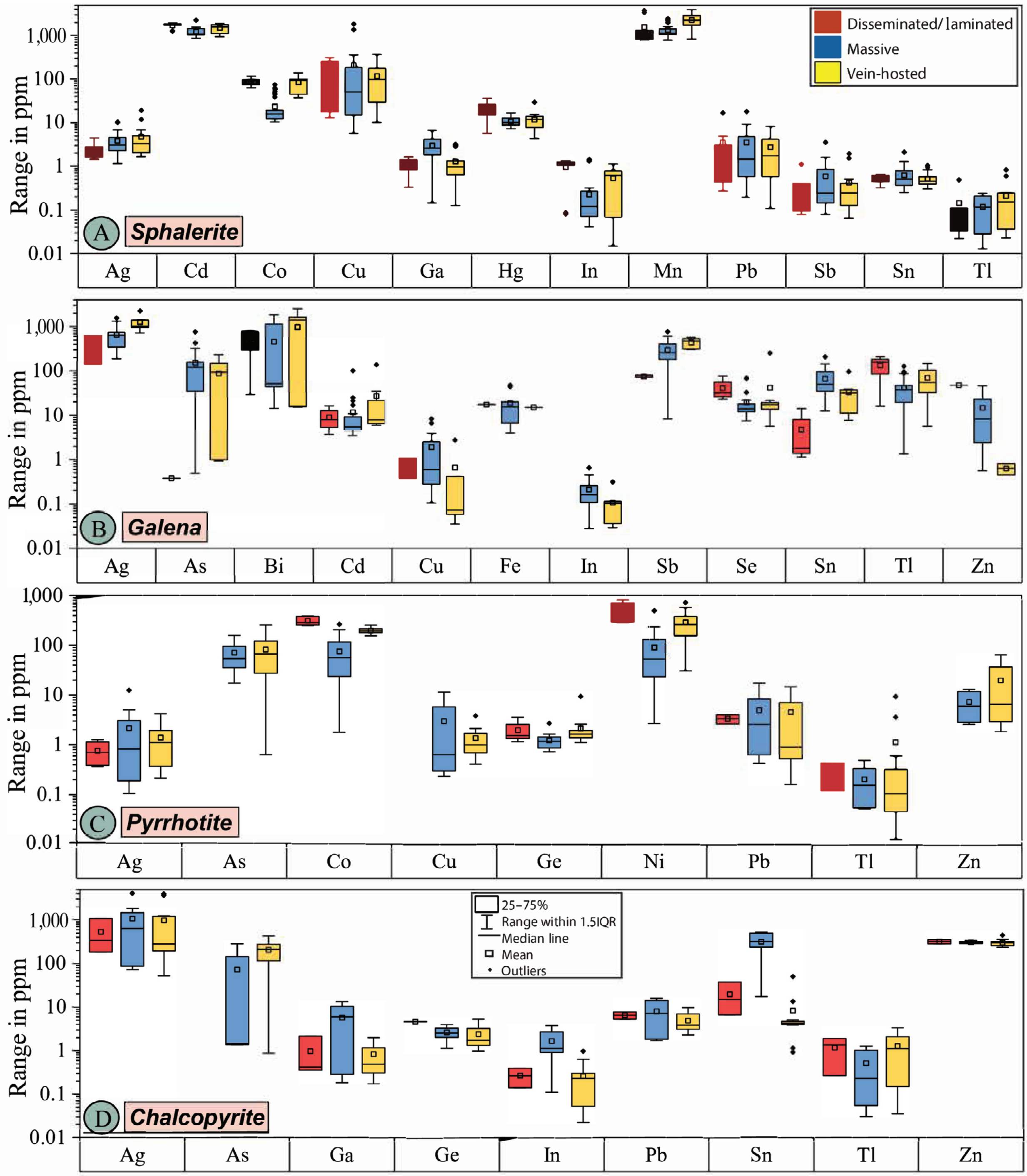

Fig. 9. Box plots of trace elements measured in A) sphalerite, B) galena, C) pyrrhotite, and D) chalcopyrite. Only those elements that are above the detection limits for most data points are selected for plotting. Box plots are constructed avoiding data pertaining to microinclusions. IQR = interquartile range.

laser ablation spot to examine the maximum trace element concentration in the sulfide structure and possible influence of microinclusions. Downhole profiles for selected spots are provided in Appendix Figure A2.

Manganese and Cd concentrations vary between 786 and 3,915 ppm and 879 and 1,925 ppm, respectively. Up to 141 pm Co was measured; however, it is distinctively lower in sphalerite from the massive ores, averaging around 23 ppm. Tin, In, and Ga are uniformly low (<10 ppm); however, some intergroup variations are noted. Indium concentrations are higher in sphalerite from the disseminated/laminated ores, whereas Ga is more abundant in the massive type. Smooth, time-resolved depth profiles of these elements suggest their presence in the structure of sphalerite. Copper concentrations span a vast range from ~6.0 ppm to 4.3 wt %, whereas Ag concentrations are <27.4 ppm. The time-resolved depth profiles of spot analyses having Cu and Ag concentrations >300 ppm and >10 ppm, respectively, show spikes in the signal, which are attributed to the ablation of chalcopyrite (as "disease") and galena inclusions, respectively. In some spot analyses of sphalerite from the massive ore, the time-resolved profiles of Cu mimic those of Ag, which is suggestive of nanoscale inclusions of Cu-Ag–bearing minerals. In many of the analyzed spots, Ag also shows ragged depth profiles that correlate with Sb, Pb, Tl, Sn, and Cu. Antimony does not exceed 11 ppm. Mercury concentrations above 16 ppm may be attributed to inclusions based on irregular time-resolved spectra. Arsenic is detected for a few grains in quartz-mica schist massive samples that contain large euhedral arsenopyrite, whereas Bi, Mo, Se, Ni, and Ge are below the detection limits.

Galena: The trace element concentrations of galena (n = 41) are summarized in Figure 9B. Average concentrations of Ag in galena from disseminated/laminated, massive, and vein-type ores are 403, 646, and 1,268 ppm, respectively. Antimony shows similar distribution patterns with average values of 73.4, 410, and 697 ppm, respectively, for the three mineralization types. Bismuth concentrations vary from 14.3 to 2,508 ppm with smooth consistent ablation depth profiles. Arsenic concentration is <1 ppm except in the arsenopyrite-bearing samples, where values as high as 760 ppm are noted. Selenium and Tl concentrations range up to 80 ppm (with an outlier value of 253 ppm) and 208 ppm, respectively, with smooth downhole ablation profiles. Indium and Sn concentrations are below 0.6 ppm and 209 ppm, respectively. Both elements show similar trends in their distribution; the lowest and highest concentrations being in disseminated/laminated and massive type ores, respectively. Highly irregular downhole ablation profiles are noted for Fe, Zn, and Cu. Cadmium, Mn, and Hg concentrations are uniformly low, whereas Ga, Ge, Co, Ni, and Mo are below detection limits.

Pyrrhotite: The trace element composition of pyrrhotite (n = 44) is summarized in Figure 9C. Most trace elements are either below detection limit or occur in low concentrations, except for Co (up to 392 ppm) and Ni (up to 820 ppm). Cobalt and Ni are comparatively depleted in the massive type, a trend observed also in sphalerite. Arsenic occurs in appreciable amounts (up to 300 ppm); however, it is below the detection limit in pyrrhotite from the disseminated/laminated ores. Copper, Zn, Cd, Pb, and Ag are mostly below 10 ppm; anomalous concentrations are attributed to the ablation of microinclusions of sphalerite or galena, or Pb-Cu sulfosalts.

Chalcopyrite: Silver concentrations in chalcopyrite display a large variation between 53 and 4,089 ppm. All spot analyses have smooth time-resolved profiles. Arsenic concentrations up to 428 ppm are noted in chalcopyrite associated with the veins; however, arsenic is low or below detection limit in chalcopyrite from the other ore types. Tin concentrations are usually below 50 ppm, except for the massive-type ores, where it increases up to 500 ppm. In a few analyses that exceed 500 ppm, the Sn signal mimics the irregular depth profiles of Zn and Pb and is therefore considered to be a result of the ablation of microinclusions of Pb-Sn sulfosalts or sphalerite. Indium and Ga concentrations are much lower (around 10 ppm) than Sn, but all three elements show a similar distribution; that is, their concentrations are highest in massive type compared to the other groups (Fig. 9D). Irregular depth profiles of Zn in analyses where the concentrations exceed 600 ppm pertain to ablation of sphalerite inclusions. Similarly, Pb concentrations exceeding 20 ppm are attributed to galena inclusions. Cobalt, Cd, Ge, Tl, Bi, Sb, and Mn are distributed uniformly with concentrations below 10 ppm, whereas Se, Mo, and Hg are below the detection limit.

Temperature

Among the temperature-sensitive elements, sphalerite from the Kayad deposit has moderate to high Mn and Fe, while being generally depleted in Ga, Ge, In, Sb, Sn, As, and Tl (in section headed "Sphalerite"). The highest temperature of 392°C, using the GGIMF thermometer, is obtained from sphalerite in the pegmatite. The disseminated-type and vein-hosted sphalerite furnish similar average temperatures of 351° ± 16°C and 350° ± 7°C, respectively, whereas the massive ores provide slightly lower temperatures (333° ± 17°C) (Fig. 10A). In the massive ore, widespread pseudomorphic replacement of löllingite by arsenopyrite is indicated by the nearly ubiquitous occurrence of löllingite relics in arsenopyrite, suggesting the latter to be a product of retrogression. The arsenopyrite contains inclusions of galena and also shares straight grain boundaries with galena and sphalerite. Using the T-X diagram of Kretschmar and Scott (1976), the minimum and maximum temperatures are constrained at 332° and 510°C from the minimum (34%) and maximum (36.2%) atomic % of As, respectively, projected on the löllingite + pyrrhotite-arsenopyrite line (Fig. 10B). The host rocks of the Kayad sulfide deposit contain fine grains of biotite defining the tectonic foliation (metamorphic biotite) in almost all the lithotypes. Coarser biotite is also observed in association with the massive and hydrothermal sulfides. For temperature calculations, pressure is assumed at 5.7 kbar as per the P-T estimates of regional metamorphism (refer to "Geologic Background" section). The metamorphic biotite as well as that associated with massive sulfides furnish temperatures between 434° and 664°C (n = 66; Fig. 10C). However, a large number of the retrieved temperatures lie between 500° and 600°C, with both the metamorphic and ore-associated biotite having overlapping ranges.

Discussion: Origin of the Sulfide Mineralization in the Kayad Deposit

SedEx mineralization

The identification of a sediment-hosted deposit is based on the lithology, depositional environment, and mineralogy of sulfides and their relationship with the host rock. SedEx deposits typically form in low-energy, euxinic environments, amongst clastic sediments and/or carbonate sedimentary rocks. Sphalerite, galena, and pyrite are the typical minerals of SedEx deposits, but pyrrhotite can be significant, especially in Proterozoic deposits such as Mt. Isa, Sullivan, Sindesar-Kalan, etc. (Deb and Pal, 2004; Leach et al., 2005). Thin bands and disseminations of sphalerite and pyrrhotite in the quartz-mica schists and quartzites of Kayad are suggestive of a sedimentary origin. Grain elongation and conformable folding of min-

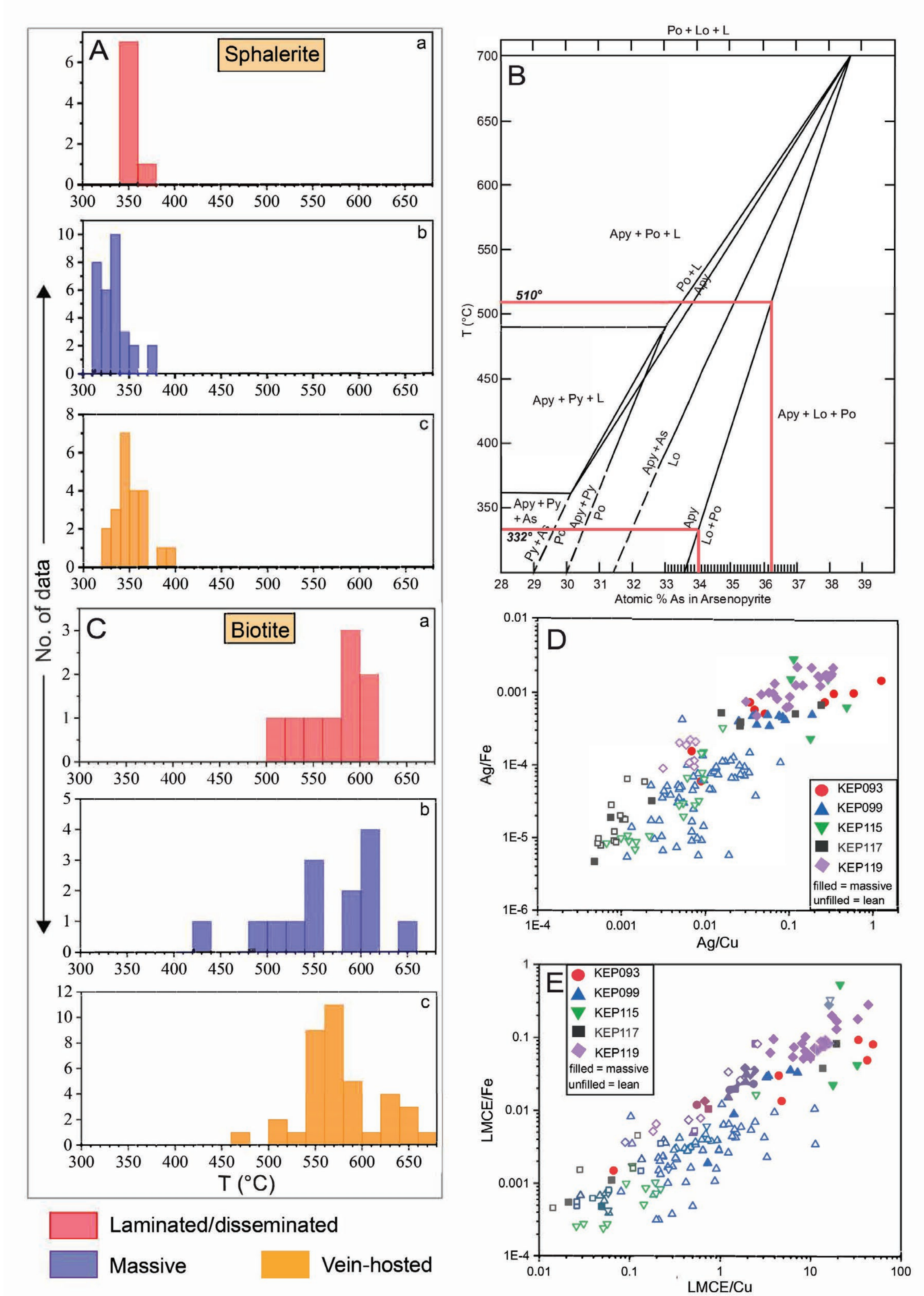

Fig. 10. A) Histogram showing temperature ranges of disseminated/laminated, massive, and vein-hosted sphalerite. B) T-X diagram of Kretschmar and Scott (1976) for calculating temperature. C) Histograms showing temperature distribution, obtained using biotite thermometry, of laminated/quartzite, massive, and vein-hosted sulfides. D) Ag/Fe vs. Ag/Cu scatter plot in massive and laminated ores. E) Low-melting point chalcophile element (LMCE)/Fe vs. LMCE/Cu scatter plot in massive and laminated ores. Different symbols represent different boreholes. Filled symbols represent massive type; unfilled symbols represent laminated type. Abbreviations: Apy = arsenopyrite , L = liquid, Lo = Loellingite, Po = Pyrrhotite, Py = pyrite.

eralized lamina with the host rock schistosity are consistent with subsequent deformation and metamorphism of the ores, which obliterated the textural and mineralogical attributes of synsedimentary hydrothermal alteration. The protolith of the graphite-bearing quartz-mica schist was likely a carbonaceous shale. The graphite-rich nature of the mineralized rocks suggests that the sulfides initially precipitated in anoxic conditions where organic matter played a role in localizing the mineralization. Furthermore, the high Zn/Pb ratio and low Cu contents are common features of primary SedEx mineralization (Shah, 2004; Leach et al., 2005), which is evident from abundant sphalerite and minor galena and chalcopyrite in the laminated ores of the Kayad deposit. Hence, we conclude that these low-grade, thinly laminated ores represent the original synsedimentary/diagenetic stratabound SedEx mineralization.

Remobilization of SedEx ore

Mobilization or remobilization of sulfides involves translocation of a preexisting disseminated or semimassive mineralization during deformation and metamorphism. This process can result in modification of the parent mineralization or creation of a new and separate daughter mineralization, which in some cases can form prospective higher-grade ores (Marshall and Gilligan, 1993; Marshall et al., 1998). The processes that drive sulfide ore remobilization are as follows: (1) solid-state dislocation flow—most sulfide minerals become ductile during deformation at lower temperatures than silicates and migrate to low-stress zones via plastic flow; (2) melt-assisted remobilization—partial melts of sulfides are generated during prograde metamorphism under suitable physicochemical conditions and mobilized to low-stress zones; (3) hydrothermal fluid-assisted dissolution and precipitation wherein fluids dissolve the sulfides and transport the elements for later deposition; or (4) a combination thereof (Marshall et al., 2000; Tomkins, 2007).

The efficiency of sulfide ore mobilization during plastic flow or sulfide anatexis is temperature dependent. Therefore, before we examine the probable causes of mobilization at Kayad, a brief discussion on the metamorphic P-T conditions experienced by the ore and host rocks during the regional tectonothermal overprint is important. Sphalerite recorded an average temperature of 350°C, which is close to its closure temperature (about 310° ± 50°C). As the rocks in and around the Kayad region have been exposed to at the least amphibolite-grade metamorphic conditions, the temperature retrieved from sphalerite cannot represent peak metamorphic conditions but rather reflects postpeak metamorphic cooling. Likewise, the arsenopyrite-löllingite pairs typify a retrograde relationship (Tomkins and Mavrogenes, 2001) and therefore the temperature obtained from arsenopyrite thermometry represents conditions during retrogression. The large spread of temperature is likely a result of variable reequilibration with decreasing temperature, possibly as a function of grain size. Nonetheless, this also implies that the peak-metamorphic temperature must have been greater than the highest retrograde temperature (510°C) recorded by arsenopyrite. The highest temperature of approx. 650°C, obtained from Ti-in-biotite thermometry, is consistent with the middle to upper amphibolite facies conditions. However, the large spread in the temperatures retrieved from biotite is suggestive of variable reequilibration during metamorphic cooling (Fig. 10C).

Remobilization by plastic flow: Textural features that characterize plastic flow have been well established by previous workers. For example, microscale to mesoscale features that indicate solid-state transformation include (1) grain-scale brittle and ductile deformational structures such as cataclasis and grain elongation, (2) discrete blebs/inclusions of a competent mineral in a continuous framework of an incompetent one or alternate bands of both, (3) folding of sulfide bands with dislocation features such as attenuation at limb and thickening at hinges, and (4) Durchbewegung texture. Macroscale features such as nose thickening and limb attenuation observed at deposit scale can provide an indication of the extent of gross remobilization (Barnes, 1987; Gilligan and Marshall, 1987; Plimer, 1987; Marshall and Gilligan, 1993). In this context, the morphology of the ores and the host rocks in Kayad is very similar to deposits like Montauban and Broken Hill; i.e., the rocks are intensely folded such that the limbs are attenuated and the dilational fold nose regions accumulate massive sphalerite-galena ± pyrrhotite ± chalcopyrite ores that disrupt the pervasive foliation in the QMS and form the bulk of the mineable lode in the Kayad (Fig. 2B). The average width of the orebodies at the limb is approx. 5 m, whereas width at the hinge is ~30 m. This is evidence of macroscale mechanical mobilization, which is reflected at both hand specimen and microscopic scale (App. Fig. A1-G). Furthermore, the presence of prominent Durchbewegung textures (Fig. 5F) wherein competent host-rock fragments occur in sphalerite mass, coarsening of massive sulfide-associated gangue minerals (Fig. 7G), and piercement veins are additional evidence of mechanical mobilization of sulfides by plastic flow, particularly of sphalerite, in Kayad deposit. Although mobilization by hydrothermal fluids can be an equally viable mechanism of ore accumulation in dilational zones, the absence of prominent hydrothermal alteration in and around the Kayad massive ores discounts extensive fluid-induced chemical mobilization.

Remobilization aided by melt generation: The idea of partial melting of sulfide as a mechanism of remobilization and ore formation, though alluded to by authors like Brett and Kullerud (1967) and Vokes (1969), garnered attention again in the early 2000s. Evidence of melting was first documented in Broken Hill, Australia, which inspired in-depth investigation across all possible melt-mobilized deposits. Textural attributes observed commonly in these deposits are (1) the presence of multiphase melt inclusions in high-temperature silicate minerals such as quartz, (2) accumulations of precious metals such as Au and Ag and sulfosalts in the remobilized ores, (3) fracture-filling by sulfides/sulfosalts, and (4) very low interfacial angles between sulfides suspected of having crystallized from a melt and restitic phases, for example, between galena/chalcopyrite/pyrrhotite and sphalerite (Frost et al., 2002, 2011).

In this context, evidence in support of melt-induced mobilization during metamorphism and deformation at Kayad can be summarized as follows:

1. Textural evidence: The massive sulfides in Kayad commonly display a typical texture where galena and chalcopyrite form cusps and carries against sphalerite (Fig. 5E). The measured dihedral angles between sphalerite-galena across different samples varies between 10.2° and 94.3°

(mean = 49.6°, σ = 20.5°) (App. Fig. A3-A, B). Such low dihedral angles suggest that the pair did not equilibrate in a solid state; rather, galena and chalcopyrite formed from a melt that wet the grain boundaries of sphalerite (Frost et al., 2011). Such a texture is ubiquitously present in the massive ores of Kayad deposit and provides prima facie evidence in favor of sulfide melting. Unequivocal textural evidence of sulfide anatexis can be best represented by quenched melt texture as demonstrated in several experimental studies. However, it is extremely difficult to quench a sulfide melt formed during metamorphism because of very slow cooling rate and consequent reequilibration. This makes it challenging to identify the presence of sulfide melts in a metamorphic rock. In the section headed "Massive sphalerite-galena ± pyrrhotite ± chalcopyrite," we have described several complex intergrowths among multiple phases such as gudmundite-pyrrhotite-galena and pyrargyrite-gudmundite-galena-pyrrhotite-chalcopyrite. The most possible mechanisms of formation of such texture are exsolution, breakdown of a metastable phase, or reequilibration of a sulfide melt during metamorphic cooling. Exsolution occurs when a phase becomes unstable in its host due to changing temperatures and commonly manifests in the form of crystallographically oriented laths or lamellae within the host phase. Previous workers (e.g., Foord and Shawe, 1989; Simanenko, 2007; Chovan et al., 2021) have reported sulfosalt exsolution textures in the form of laths, lamellae, or acicular grains within larger galena grains. The intergrowth textures shown in Figures 5H and 7A-B do not exhibit typical exsolution features. It is, however, possible that the exsolved phases may have been expelled from the host mineral during subsequent recrystallization. Such a process will likely result in simpler textures, such as accumulation of the exsolved phase along the grain boundaries of the parental grain (such as Fig. 7C-E), and is unlikely to form a complex intergrowth of multiple phases around the parental grain (e.g., Fig. 5H; intergrowth of multiple phases at the boundary of galena). Therefore, it is difficult to explain these intergrowths by simple exsolution (e.g., Fig. 7A). Even some apparently simple intergrowth textures such as gudmundite-sphalerite shown in Figure 7B cannot be explained by exsolution. For example, in this case, sphalerite must contain significant dissolved Sb to form nearly equal volume of exsolved gudmundite, but high proportions of Sb have rarely been reported in sphalerite (Cook et al., 2009). Moreover, sphalerite (with or without associated gudmundite) in Kayad has maximum Sb concentrations of 2 ppm, which is insufficient to form the proportion of gudmundite seen in the intergrowths. Alternatively, to ascertain whether the intergrowths formed due to destabilization of a metastable phase, the bulk compositions of some selected intergrowths (gudmundite-pyrrhotite-galena and pyrargyrite-gudmundite-galena-pyrrhotite-chalcopyrite; Figs. 5H, 7A) were calculated using the volume proportion and elemental compositions of the constituent minerals. The calculated compositions are dominated by Fe, Sb, and Pb with minor amounts of Cu and Ag and can be best represented by PbS-FeS-Sb_2S_3 ternary system. The only known sulfosalt that could provide the necessary elements for these intergrowths through breakdown of a single metastable phase is jamesonite ($Pb_4FeSb_6S_{14}$). The deduced compositions, plotted in the PbS-FeS-Sb_2S_3 ternary diagram proposed by Chang and Knowles (1977), fell well outside the jamesonite stability field, thereby precluding jamesonite as a precursor phase. Therefore, we argue that there is a possibility for these complex intergrowths to be reequilibrated/recrystallized melt. Nevertheless, we do not discount the possibility that some of the mineralogically simple intergrowths, such as galena and Ag tetrahedrite (Fig. 7E-F; App. Fig. A1-B, E, F), and some of the inferred inclusions of sulfosalts in the sulfides can potentially be exsolutions.

2. Mineralogical and chemical: As discussed above and earlier in the section headed "Massive sphalerite-galena ± pyrrhotite ± chalcopyrite," several aggregates of sulfosalts in association with the massive ores are identified in samples of various mine levels, which suggests their widespread occurrence. Time-resolved ablation signals show spikes in certain elements such as Ag (>3,000 ppm) and Sb (>600) in galena (App. Fig. A2) in massive ores that indicate the presence of microinclusions of Ag-antimonides or sulfo-antimonides (Palero-Fernández and Martín-Izard, 2005). Similarly, the covariation of Ag, Pb, Tl, and Sb concentrations in the time-resolved ablation spectra of many sphalerite spot analyses suggests ablation of nanoinclusions of sulfosalts such as pyrargyrite (which also occurs as a separate phase in the massive ore, refer to section above), boulangerite, and rayite, all of which have been reported from the remobilized ores in the Zn-Pb deposits of the Aravalli-Delhi fold belt. These nanoinclusions may be the result of exsolution of sulfosalt phases from the major sulfides during metamorphic cooling.

The ubiquitous presence of sulfosalts in massive ores provides prima facie evidence that the massive ores are enriched in Ag + Bi + As + Tl + Sb. These elements are collectively known as low melting chalcophile elements (LMCEs) as they can depress the melting temperatures of common sulfide minerals (Frost et al., 2002). To further examine the relative enrichment of LMCEs in the massive ore, bulk concentration ratios of major elements and LMCEs (Ag + Bi + As + Tl + Sb) from massive mobilized ore and laminated SedEx ore horizons intercepted in different boreholes were plotted (Figs. 10 D-E; App. A3-D, E). The Ag/Fe vs. Ag/Cu and LMCE/Fe vs. LMCE/Cu plots show a relative enrichment of Ag and the LMCEs over Cu and Fe in the massive ore. This is also justified by the fact that Ag is recovered as a by-product of mining in Kayad. These observations provide evidence for the preferential partitioning of the LMCEs into the melt during sulfide anatexis, which contributed to the massive ore. However, Ag/Pb ratios of the two types show overlap. We propose that substantial removal of galena from the laminated SedEx ore (indicated by rarity of galena in the restitic ores) by melting (low dihedral angle between galena and sphalerite in the massive ore) led to limited fractionation of Ag from Pb (since galena is the primary host of Ag) between melt and restite, leading to the overlap between the lean and massive ores. Similarly, LMCE/Fe vs. LMCE/Zn shows an interesting distribution where LMCE/Zn values are highly variable in the massive ore and overlap with the lean ores. This pattern may also be attributed to the differential response of mineral species to the processes involved. Notably, galena and chalcopyrite appear to exhibit higher susceptibility to melting, whereas sphalerite appears to be comparatively more influenced by plastic flow, which is unlikely to fractionate Zn and therefore will not lead to any systematic change in elemental ratios involving Zn between restite and the mobilized ores. The textural, mineralogical, and geochemical evidence thus collectively suggest that galena and chalcopyrite were most likely mobilized dominantly by sulfide anatexis, and sphalerite was mobilized primarily by plastic flow.

Having discussed the evidence of melting, we now assess whether the metamorphic condition was suitable for such melt generations at Kayad. We have discussed above that some of the sulfide-sulfosalt intergrowths may represent reequilibrated melt. Presence of LMCEs is critical to the melt-initiation process since the LMCE-bearing sulfosalts start melting at significantly low temperatures. For example, the most common sulfosalts in Kayad, namely pyrargyrite and gudmundite, can be represented in the Ag-Sb-S and Fe-Sb-S systems, respectively, wherein the first Sb-rich melt in the Ag-Sb-S system forms at 450°C and all ternary phases melt by 510°C (Frost et al., 2002). Melting of pyrargyrite occurs at 485°C at 1 atm (Frost et al., 2002). The first melt in the Fe-Sb-S system is shown to appear at 545°C through the reaction of pyrite and stibnite; however, no experiment to synthesize gudmundite has been successful to date (Frost et al., 2002). Clark (1966) demonstrated that As-free gudmundite disintegrates at 280°±10°C when heated in vacuum. Therefore, we think that the inferred reequilibrated melt may form by partial melting during amphibolite-grade metamorphism. Govindarao et al. (2017) conducted experiments in the Ni-Sb system and established that Ag greatly depresses the melting point of the Ni-Sb system and it is possible to generate an Ag-Sb-Bi melt at 680°C. Presence of elements such as Pb and Cu can further depress the melting point. Removal of Ag through formation of Ag-bearing phases such as $CuPbSbS_3$. Ag_2S, $2FeS.Ag_2S$, $3PbS.Ag_2S$, etc., enriches the residual melt in Ni and Sb, which consequently forms breithauptite- and nisbite-like phases. They also demonstrated that in the ZnS-PbS-FeS-Cu_2S-Sb_2S_3-As_2S_3 system, melting is expedited by PbS and As to occur at 500°C, which is manifested as intergrowths of galena with tetrahedrite in various proportions (Govindarao et al., 2020). The presence of breithauptite and intergrowths of galena with Ag-bearing tetrahedrite in Kayad, therefore, may suggest the possibility of a sulfosalt melt at temperatures of 600°C. Partitioning of precious metals and LMCEs and presence of LMCE-bearing sulfosalts/minerals occasionally with complex intergrowth textures (interpreted as reequilibrated melt; "Textural evidence" in the section headed "Remobilization aided by melt generation") in massive ores metamorphosed to the middle/upper amphibolite facies is consistent with sulfide melt mobilization, which tends to fractionate melts toward enrichment in LMCEs (Tomkins et al., 2007).

Below we address the possibility of melting and mobilization of the major sulfide minerals. Since the massive ore in Kayad comprises sphalerite + galena ± chalcopyrite ± pyrrhotite, experimental studies involving these minerals are relevant. Experiments at sulfur fugacity that favors the stability of pyrrhotite instead of pyrite show that melting can occur at 830°C in the PbS-FeS-ZnS system (Mavrogenes et al. 2001) and at 730°C if $CuFeS_2$ is incorporated (Stevens et al., 2005). The eutectic temperature in the PbS-FeS-ZnS system can be further depressed by about 30°C if 1 wt % Ag_2S is added to the system (Mavrogenes et al., 2001). However, the cumulative effect of Cu, Ag, and other LMCEs on the depression of melting temperature is not known. Pruseth et al. (2014) demonstrated that melting is possible at 600°C under high-sulfur-fugacity conditions to generate a ZnS-rich melt in the ZnS-PbS-FeS-Cu_2S system. From these experimental studies, it appears that sulfide melts ("true melt"; Mavrogenes et al., 2013) from which common minerals such as pyrite, pyrrhotite, chalcopyrite, galena, and sphalerite can crystallize may form during amphibolite facies metamorphism if the S-fugacity is sufficiently high, whereas substantially higher temperature is required to generate such melt in low-S–fugacity conditions. During prograde metamorphism, desulfidation of pyrite to pyrrhotite increases S-fugacity (Tomkins, 2007) and may thus promote melting. The rarity of pyrite in Kayad is apparently at odds with experiments and deposits where significant melting of common sulfides has been demonstrated. In apparent similarity with Kayad, the Sindesar-Khurd deposit in the Aravalli-Delhi fold belt is also dominated by pyrrhotite, and the peak metamorphic temperature is inferred to be 580° to 600°C (Govindarao et al., 2017). Integrating experimental studies and field investigation, the authors demonstrated melt-assisted sulfide mobilization at Sindesar-Khurd despite being metamorphosed to relatively lower temperatures, citing pseudomorphs of pyrrhotite after pyrite as evidence of desulfidation during metamorphism. Though pyrrhotite is the most abundant iron sulfide in Kayad, irregular pyrite relics are seen within large masses of pyrrhotite (Fig. 5C), which may be construed as evidence of extreme desulfidation of pyrite to release S, perhaps allowing sulfide melting during metamorphism. Pruseth et al. (2014) showed that even in initial sulfur excess conditions, the melt generated will always be sulfur deficient. Furthermore, the original metal inventory of the SedEx ore should have been enriched in LMCEs as suggested by abundance of the sulfosalts and minerals in the mobilized massive ore, which might have facilitated melt generation by lowering the melting temperature even further. Once melting is initiated, LMCEs present as traces in sulfide structure could have diffused into the melt, thus expediting the mechanism (Frost et al., 2002). The higher Ag/Fe, Ag/Cu, LMCE/Fe, and LMCE/Cu in the massive ores compared to the lean ores also support this mechanism. We have noted localized fluid-induced alteration around massive sulfides where pyrrhotite and chalcopyrite dominate, suggesting the presence of fluid in the melt or during melting. Wykes and Mavrogenes (2005) have experimentally shown that at 1.5 GPa, H_2O dissolved in melt can depress the eutectic temperature in the PbS-ZnS-FeS system from 900° to 865°C. Therefore, the participation of fluids could have lowered the melting temperature, although we envision that its effect was minimal due to lower pressures (approx. 0.6 GPa) during regional metamorphism. Fluids might have played a more important role in lowering the viscosity of the sulfide melt, thereby facilitating its migration.

Remobilization assisted by fluid: Unlike the massive ores, the vein-hosted Fe-Cu ± Zn ± Pb mineralization at Kayad comprising pyrrhotite and chalcopyrite is ubiquitously associated with prominent hydrothermal alteration zones. Replacement of oligoclase by orthoclase + albite (Figs. 6A, E, F), orthoclase + chamosite ± Al-pumpellyite (Figs. 6G, 8D), replacement of muscovite by K-feldspar + albite + chamosite + Fe-rich muscovite (Fig. 6H), and the preponderance of coarse-grained secondary biotite along the vein wall are evidence of K ± Na ± Fe metasomatism. Plagioclase alteration along grain boundaries and cleavages marks the infiltration paths of K-Na–bearing fluids, which most likely coprecipitated orthoclase and albite as suggested by their mosaic-like texture (Nurdiana et al., 2021). Moreover, the crowding of pores in zones of secondary albite is consistent with fluid-driven dissolution-reprecipitation (Putnis, 2009). Common replacement of K-feldspar by allanite + epidote + monazite in association with Fe-Cu sulfides suggests calcic alteration by a separate pulse of fluid that postdated earlier potassic alteration.

Zn-Pb ± Fe ± Cu sulfide mineralization in discordant K-feldspar veins in quartz-mica schist suggests that Zn-Pb mineralization was synchronous with K-metasomatism, similar to the hydrothermal Fe-Cu ± Zn ± Pb mineralization. The replacement of the vein K-feldspar by prehnite, pumpellyite and albite with traces of clinochlore (Figs. 8F, H) suggests an influx of Ca-Na-rich fluid overprinting an earlier K-metasomatized assemblage. Similar to hydrothermal Fe-Cu sulfide mineralization, replacement of K-feldspar by allanite + epidote + monazite suggests an event of Ca-metasomatism following K-metasomatism. Thus, similar hydrothermal alterations are associated with both Fe-Cu sulfide and Zn-Pb mineralization. Overall, two disparate hydrothermal events can be inferred wherein an early K ± Na ± Fe alteration event was overprinted by a later Ca ± Na alteration event, both being associated with sulfide remobilization.

Mineralization in calcsilicate is sparse. The association of calcite, albite, quartz, epidote, titanite, hornblende, and chlorite likely indicates that the pyrrhotite and chalcopyrite veinlets formed during Ca-Na alteration of the host rock.

Conclusions

Based on sulfide mineral assemblages, mode of occurrence, microtextures and associated alteration features, three mineralization styles are recognized at Kayad. The disseminated/laminated type in quartz-mica schist and quartzite is conformable and concurrently deformed with the host-rock fabric and constitutes the primary, synsedimentary/diagenetic SedEx mineralization. This was followed by synmetamorphic and syndeformational remobilization of SedEx ores to dilational zones of regional fold structures thus forming the massive-type Zn-Pb resource at Kayad. Key diagnostic features include distinctive textural and mineralogical clues such as durchbewegung texture, low dihedral angles among sulfide phases, presence of LMCE-bearing sulfosalts forming complex intergrowth textures (possibly originating from a former melt), and abundant exsolutions/inclusions of Ag-Sb-Cu-Tl phases in massive sulfides. Additionally, lack of significant hydrothermal alteration implies solid-state mobilization paired with partial melting of preexisting sulfides. However, further quantification of the relative importance of the two processes is required to comment on their contribution in forming large-scale deposits. The vein-hosted ores characterized by intense Fe-Cu mineralization with minor Zn-Pb sulfide in the footwall side of the massive ores are of hydrothermal origin. Their alteration footprints are suggestive of a pervasive K + Na ± Fe alteration and an overprinting subsidiary Ca ± Na alteration, which occurred concurrently with the melt-induced mobilization during regional metamorphism. Hence, the ore mineralization in Kayad is a result of a mixed mobilization process, which has significantly enhanced the economic status of the deposit by garnering substantial concentrations of ore and precious metals (such as Ag).

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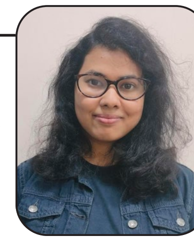

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